

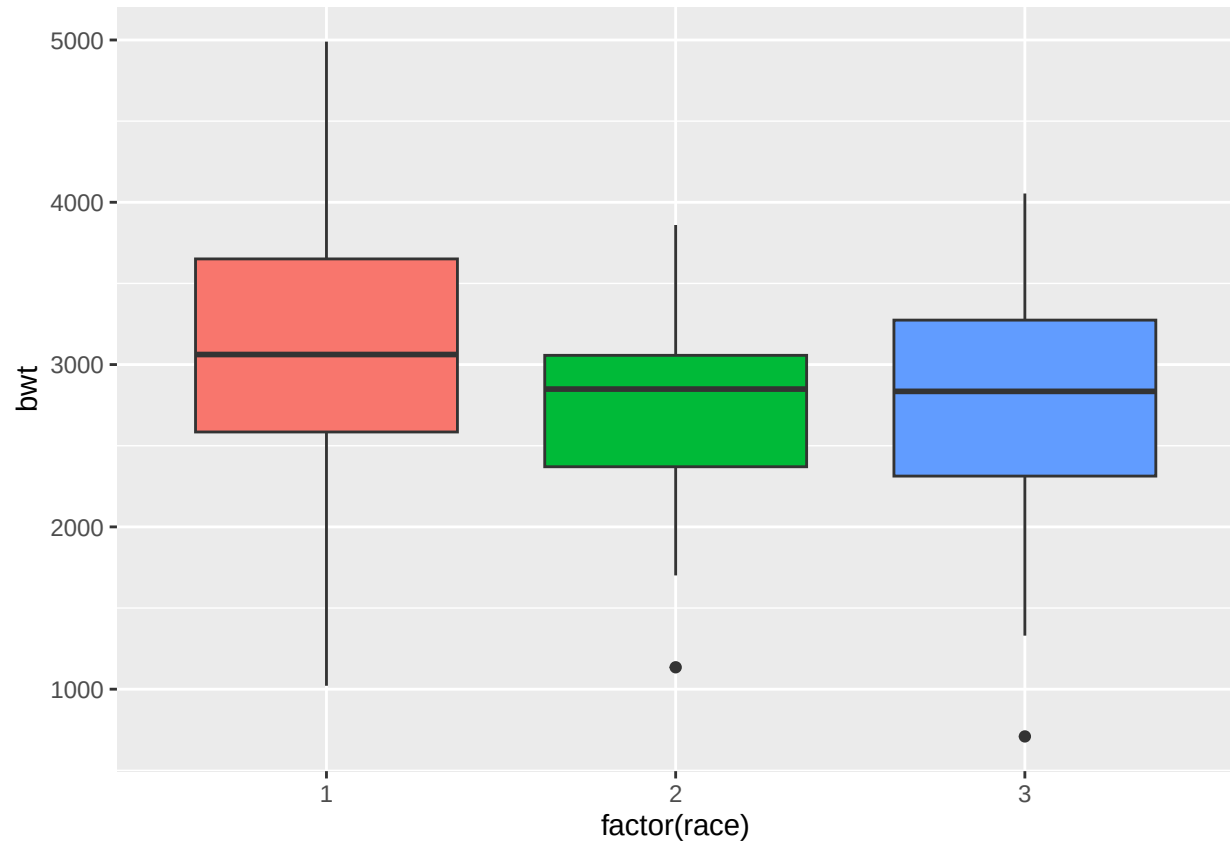
ggplot Functions for Finishing Purposes

Noratiqah Mohd Ariff

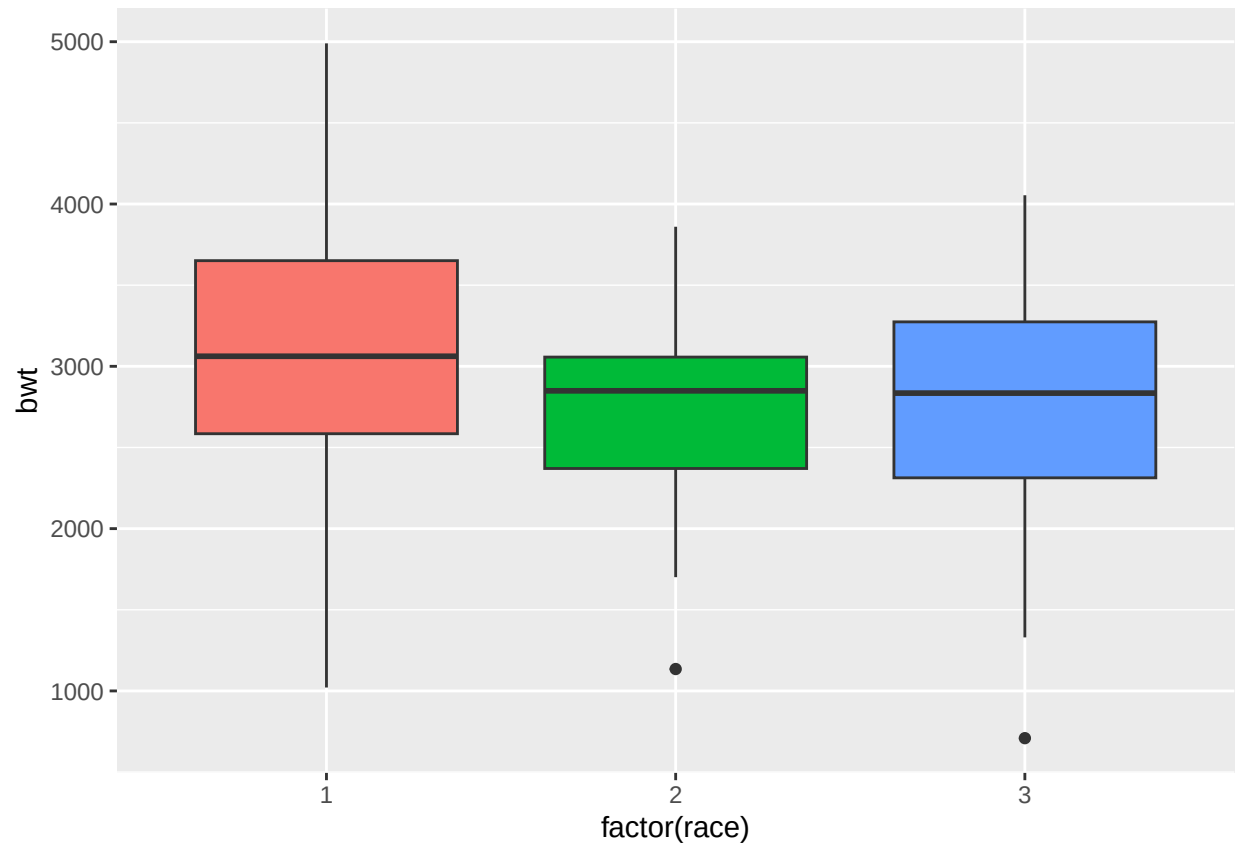
Legend

Remove Legend

```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt,fill=factor(race)))  
gg+geom_boxplot()+guides(fill=FALSE)
```

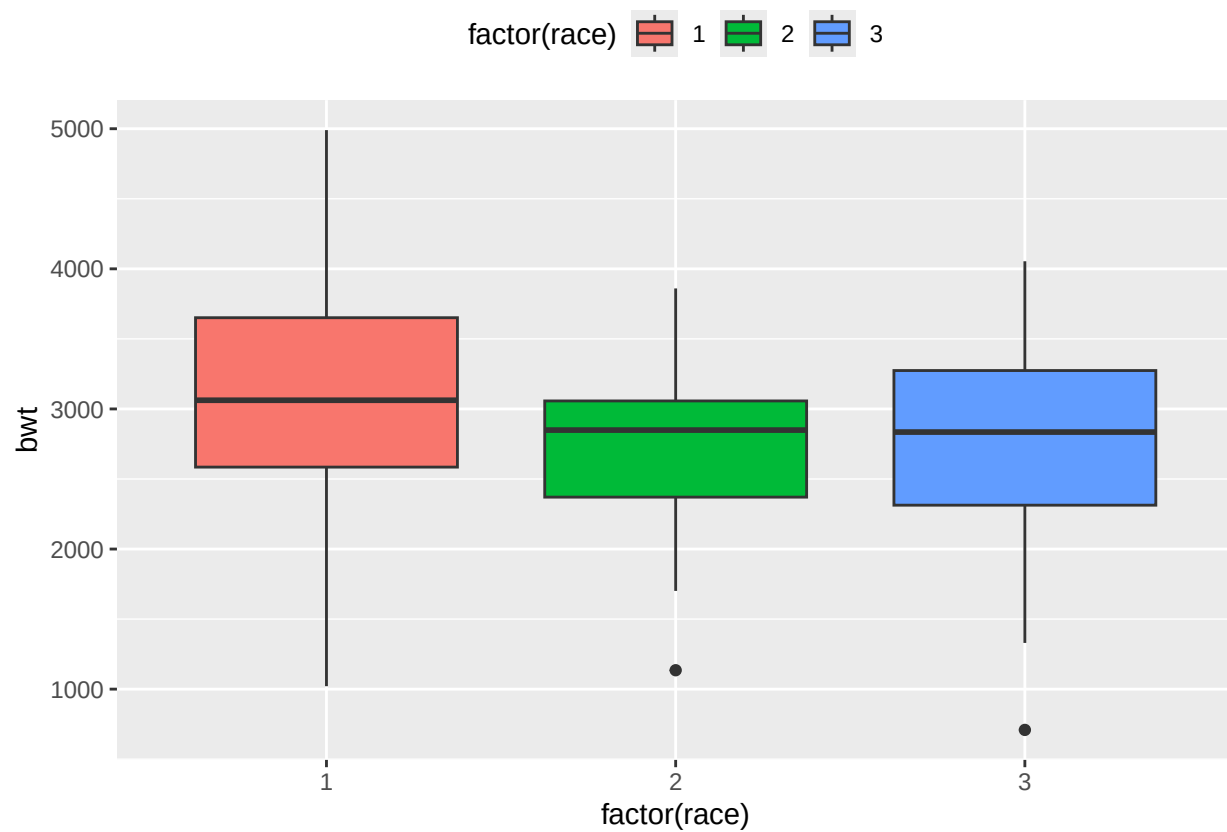


```
gg+geom_boxplot()+scale_fill_discrete(guide=FALSE)
```

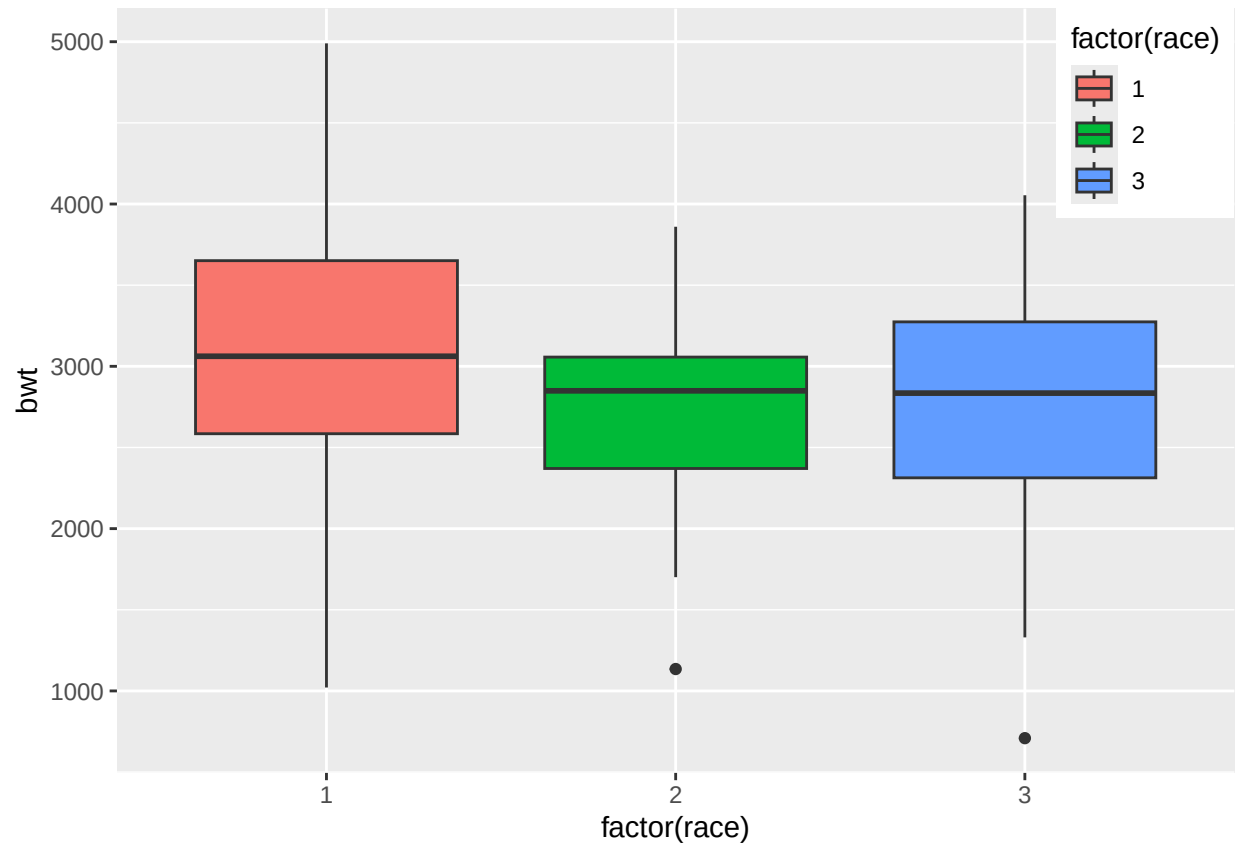


Change Position

```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt,fill=factor(race)))  
gg+geom_boxplot()+theme(legend.position="top")
```

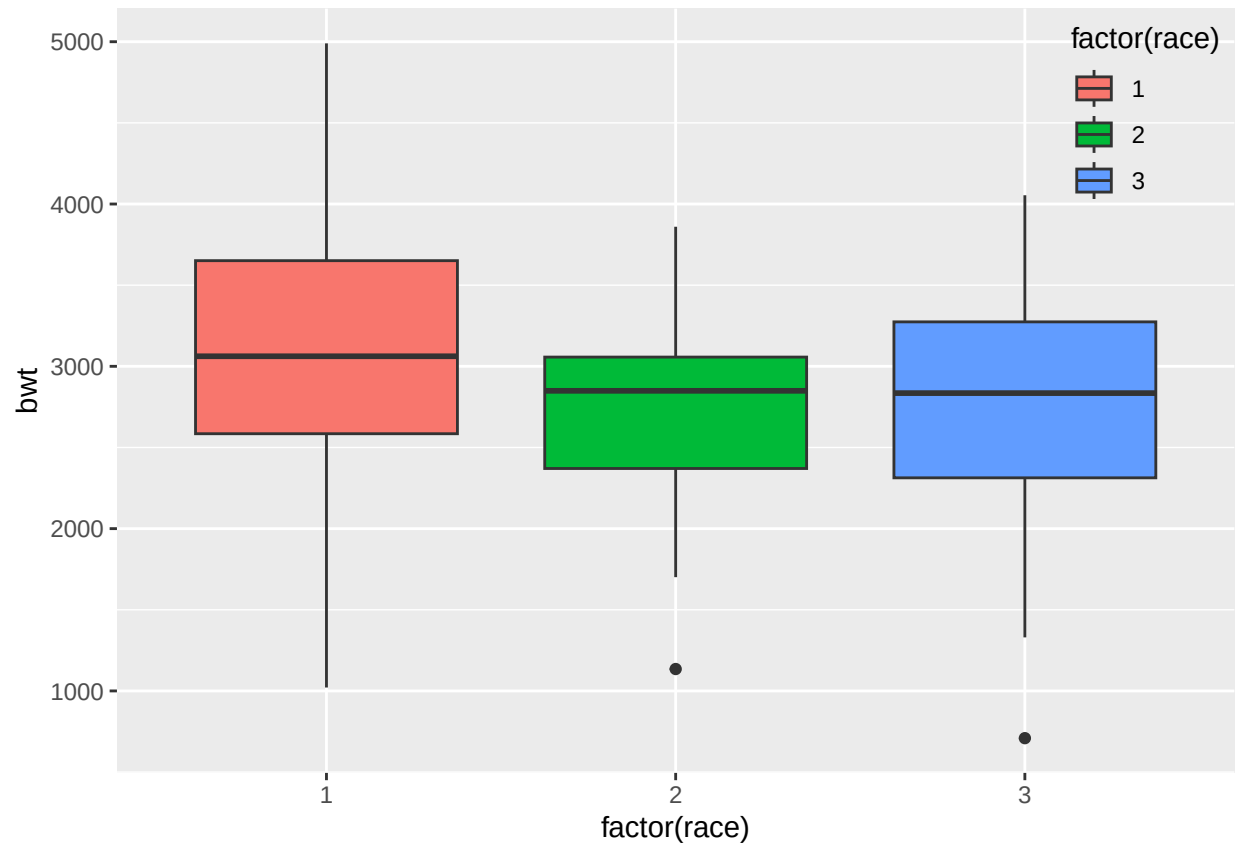


```
gg+geom_boxplot()+theme(legend.position=c(1,1),legend.justification=c(1,1))
```



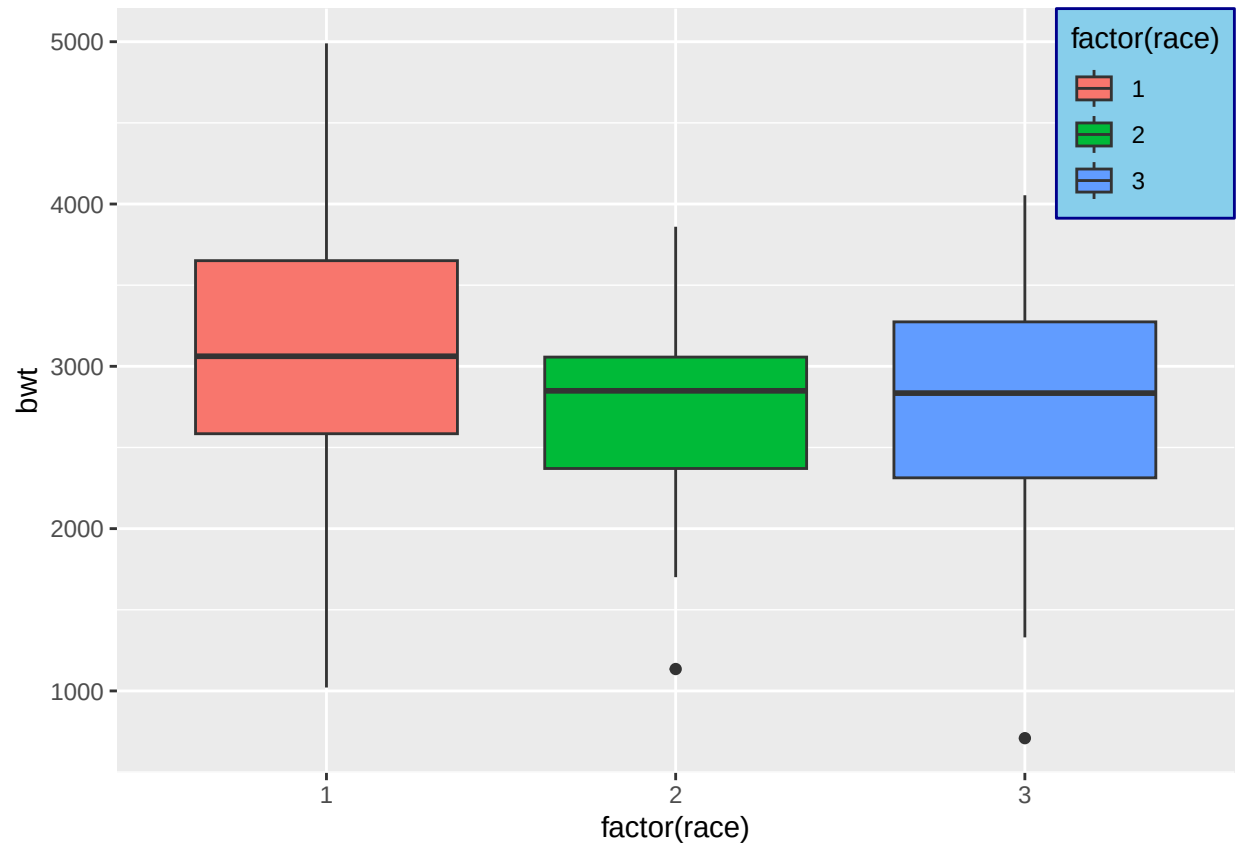
Remove Background

```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt,fill=factor(race)))
gg+geom_boxplot()+theme(legend.position=c(1,1),
legend.justification=c(1,1))+
theme(legend.background=element_blank())+
theme(legend.key=element_blank())
```



Change Background

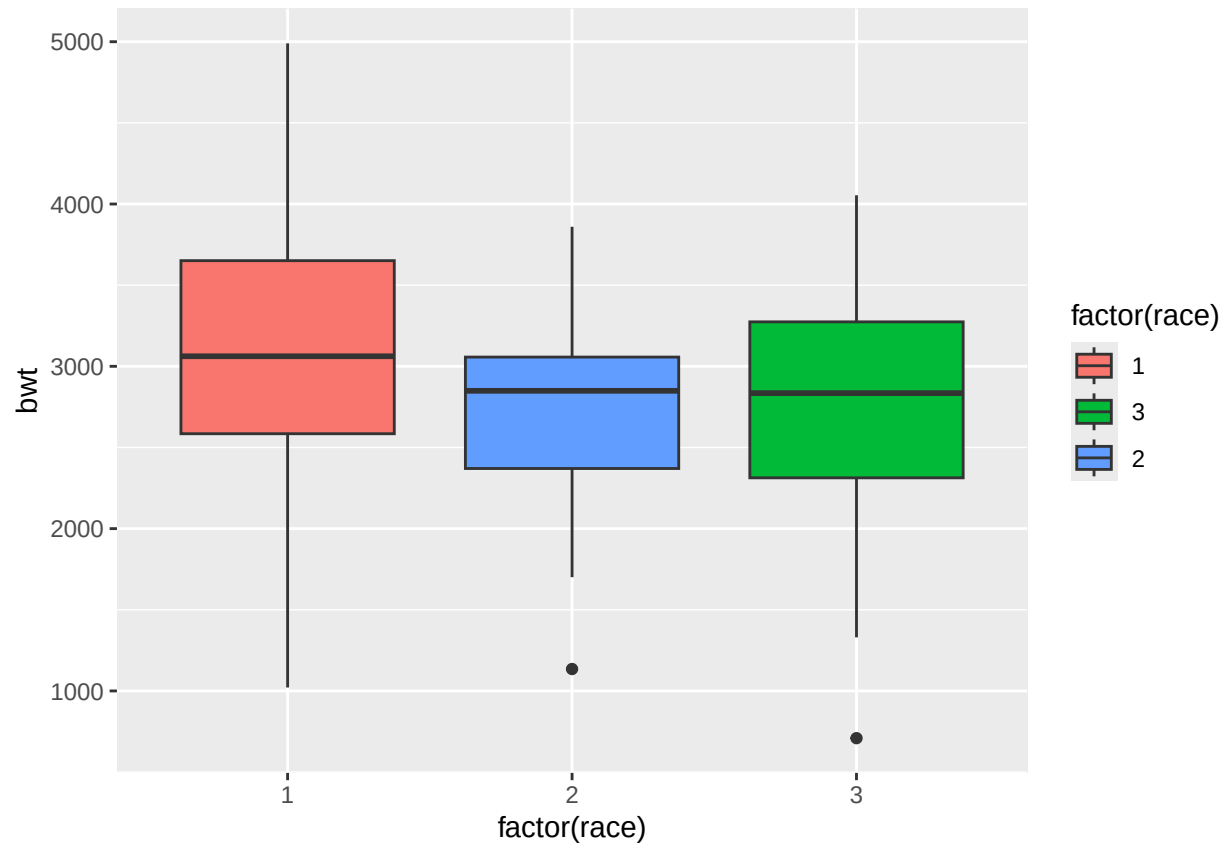
```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt,fill=factor(race)))
gg+geom_boxplot()+theme(legend.position=c(1,1),
legend.justification=c(1,1))+
theme(legend.background=
element_rect(fill="skyblue",colour="darkblue"))+
theme(legend.key=element_rect(fill="skyblue",colour="skyblue"))
```



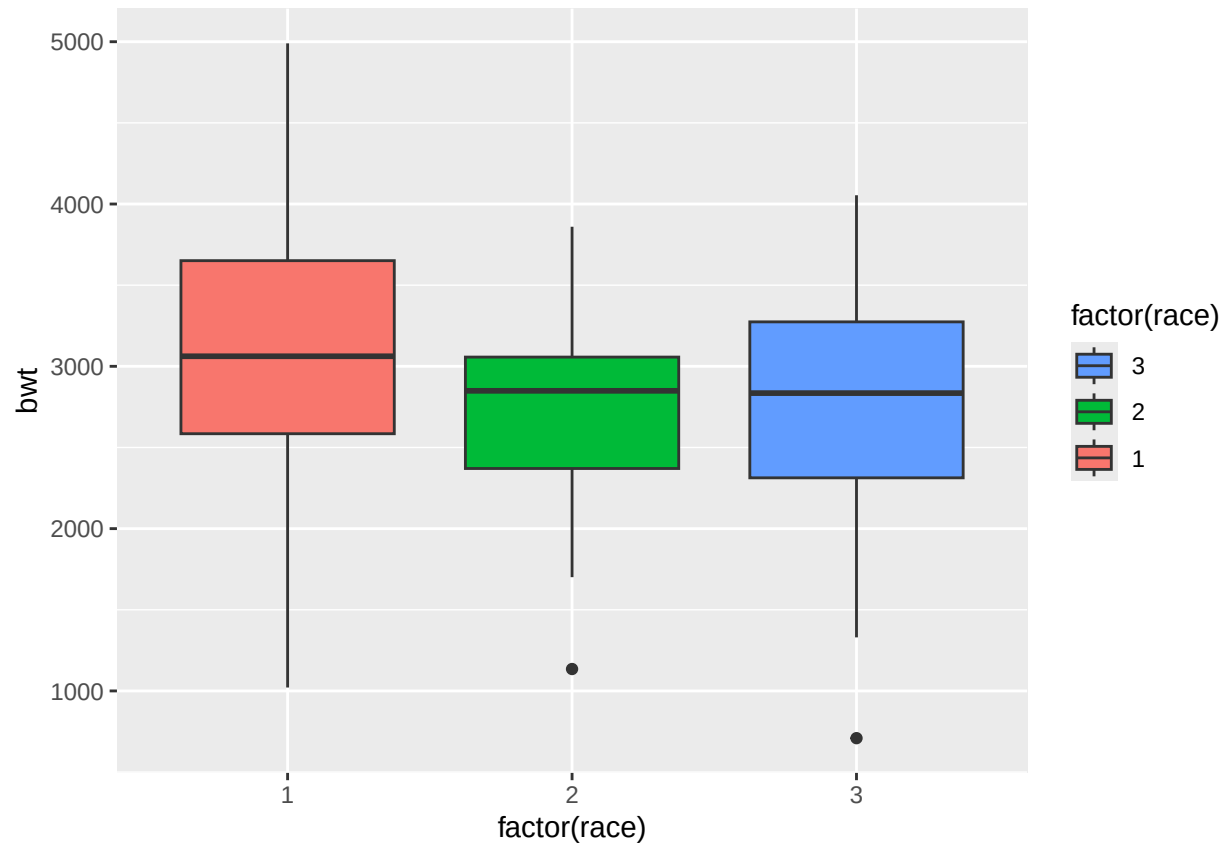
Legend

Order Items

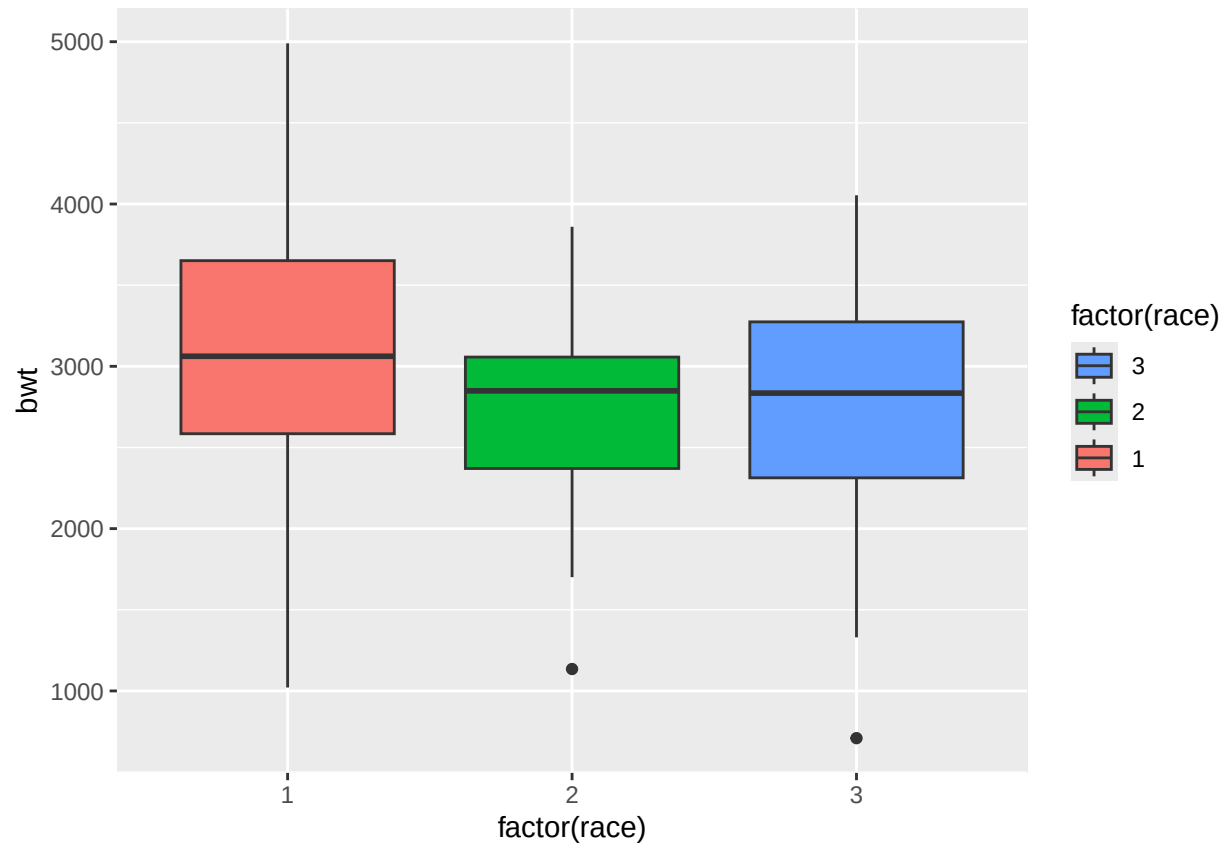
```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt,fill=factor(race)))  
gg+geom_boxplot()+  
scale_fill_discrete(limits=as.character(c(1,3,2)))
```



```
gg+geom_boxplot()+guides(fill=guide_legend(reverse=TRUE))
```

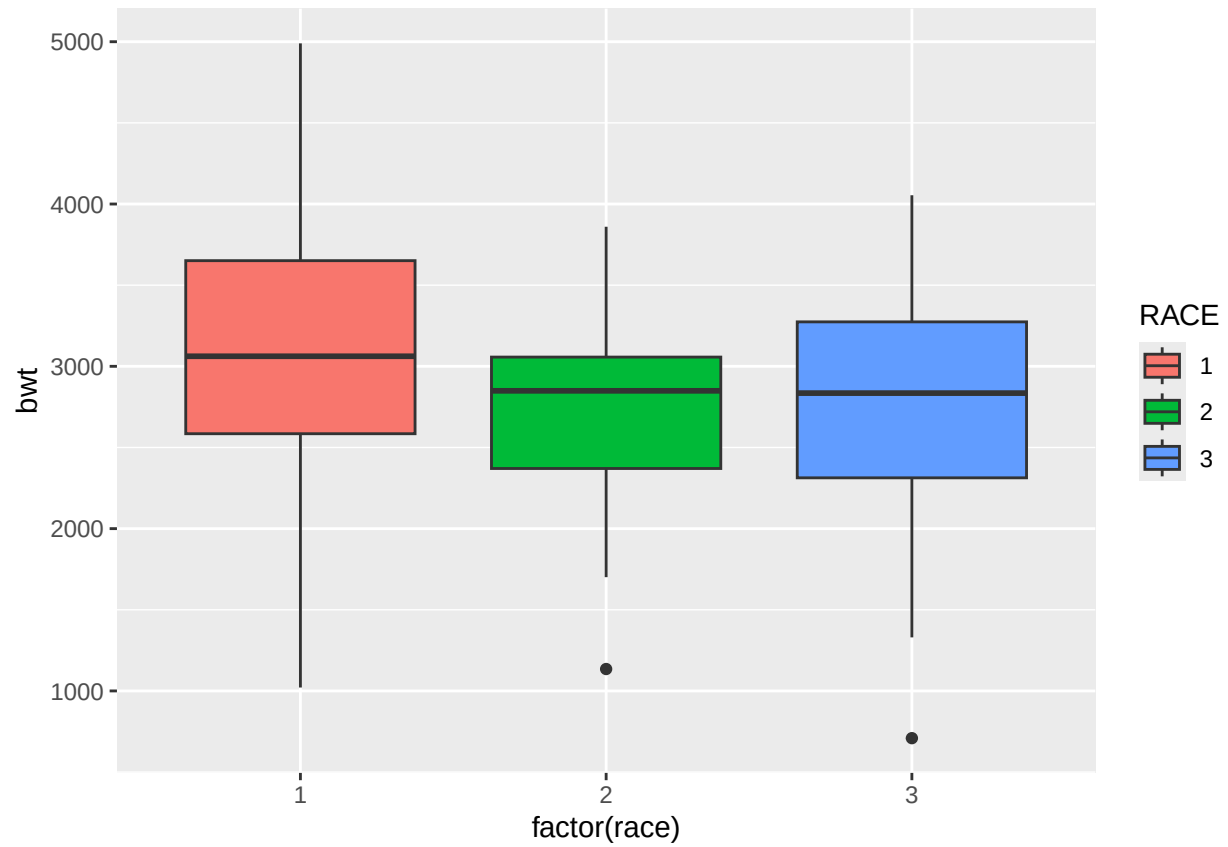


```
gg+geom_boxplot()+scale_fill_hue(guide=guide_legend(reverse=TRUE))
```

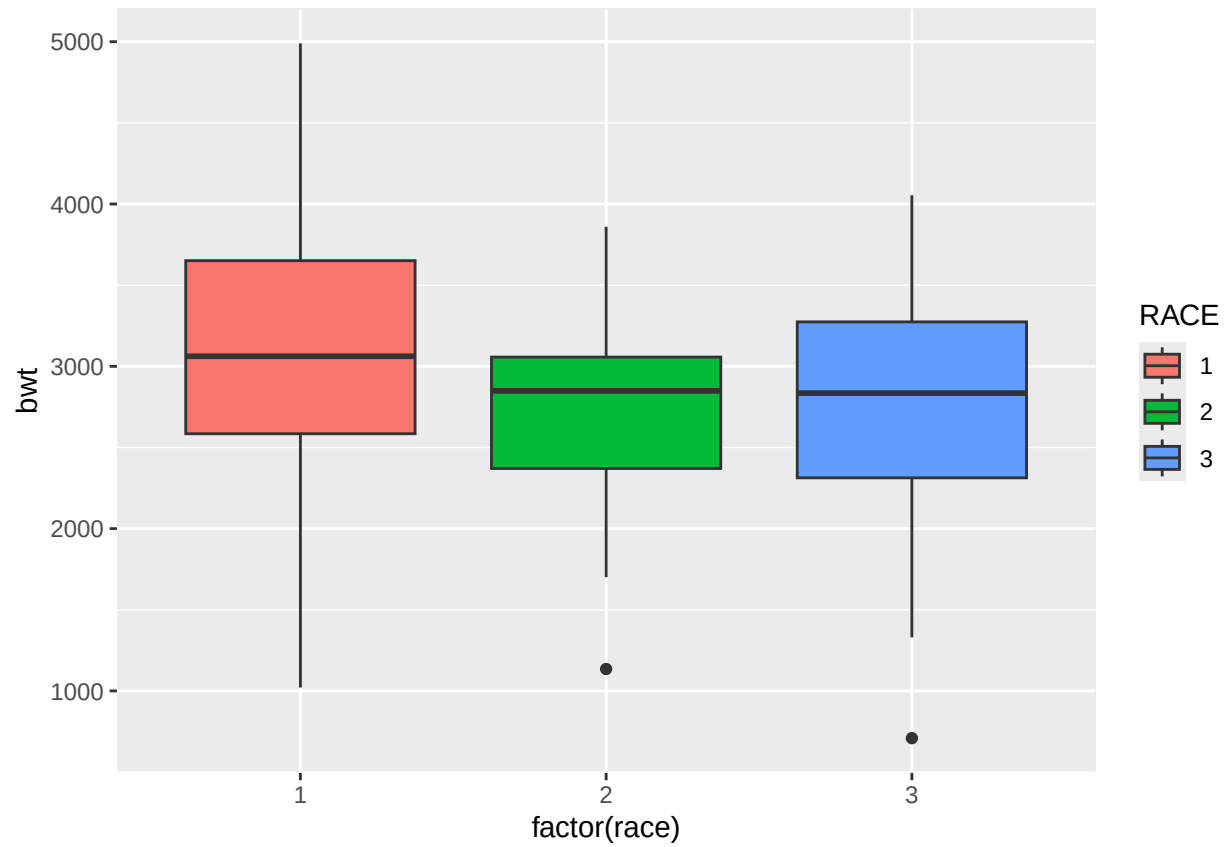



Title

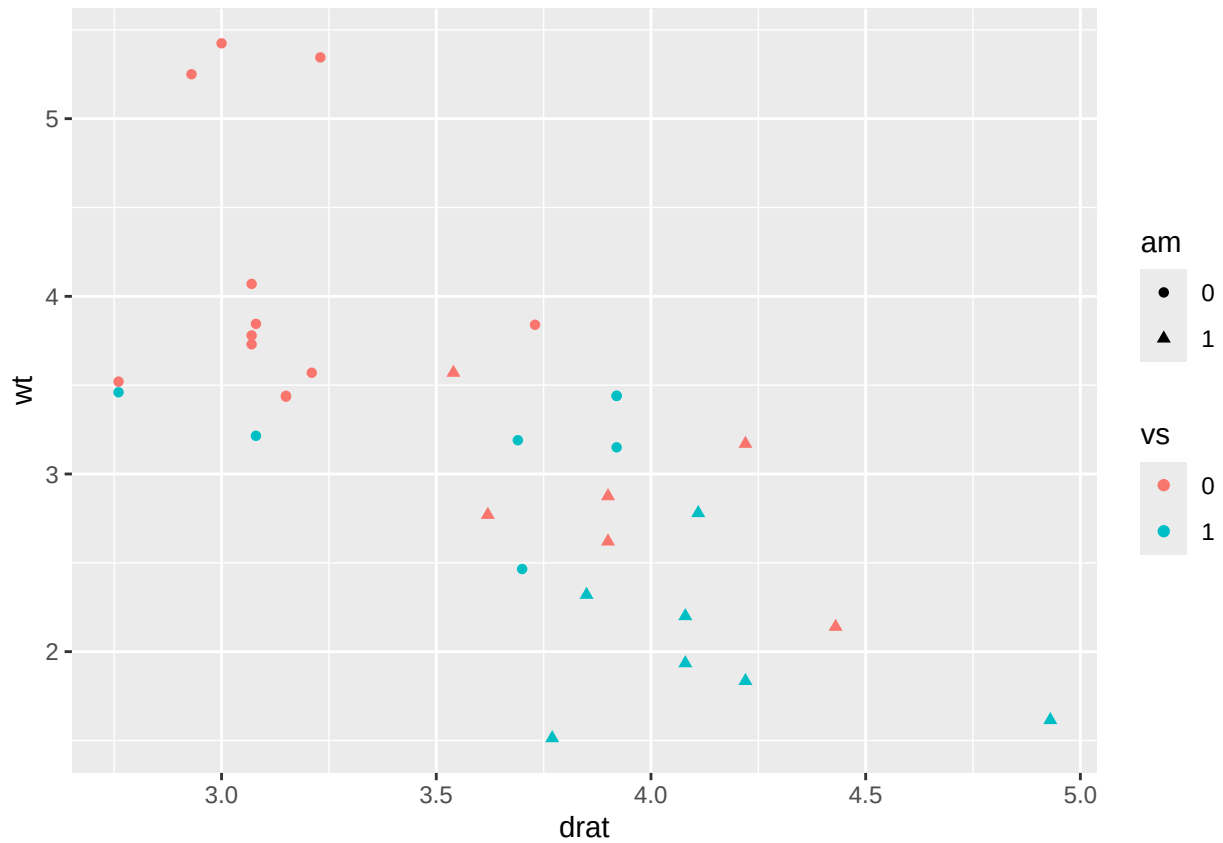
```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt,fill=factor(race)))  
gg+geom_boxplot()+labs(fill="RACE")
```



```
gg+geom_boxplot()+scale_fill_discrete(name="RACE")
```



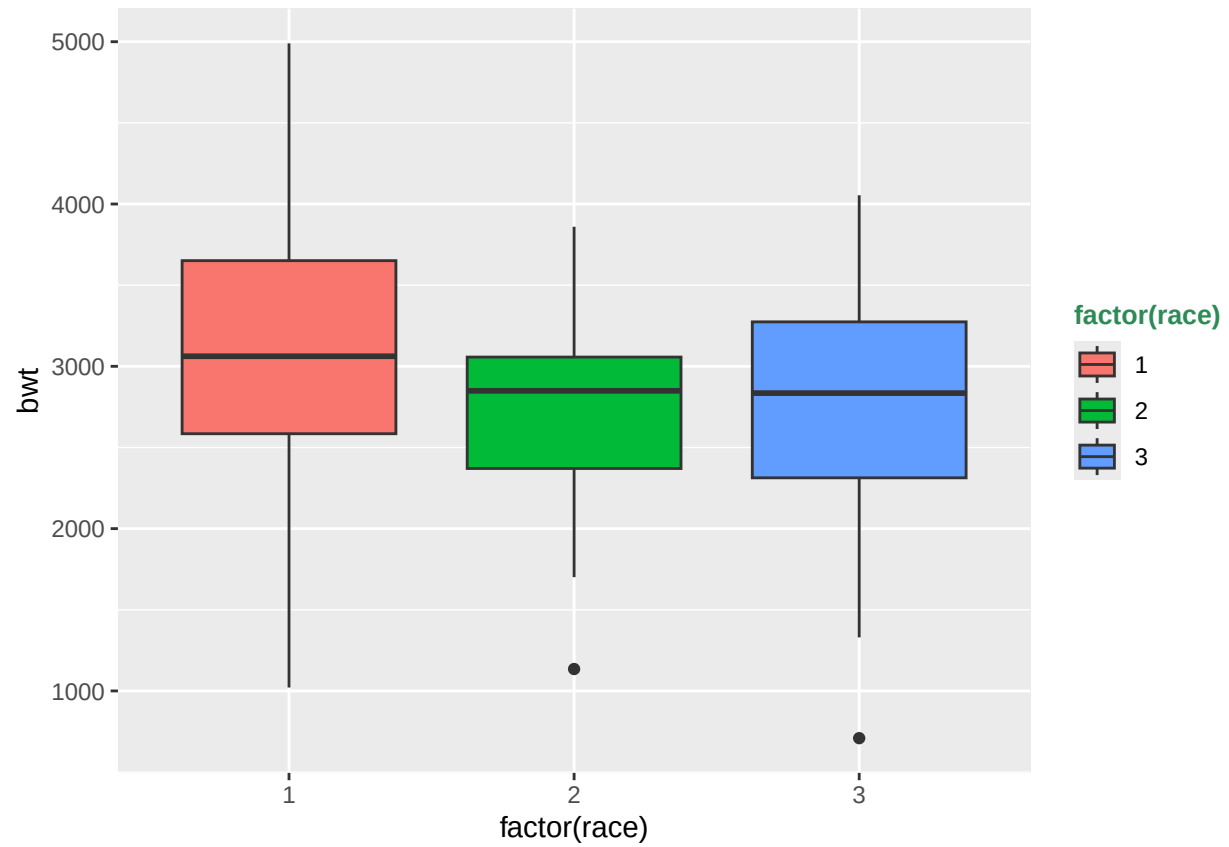
```
ggplot(mtcars, aes(x=drat, y=wt, shape=as.factor(am),  
  colour=as.factor(vs)))+  
geom_point()+labs(shape="am", colour="vs")
```



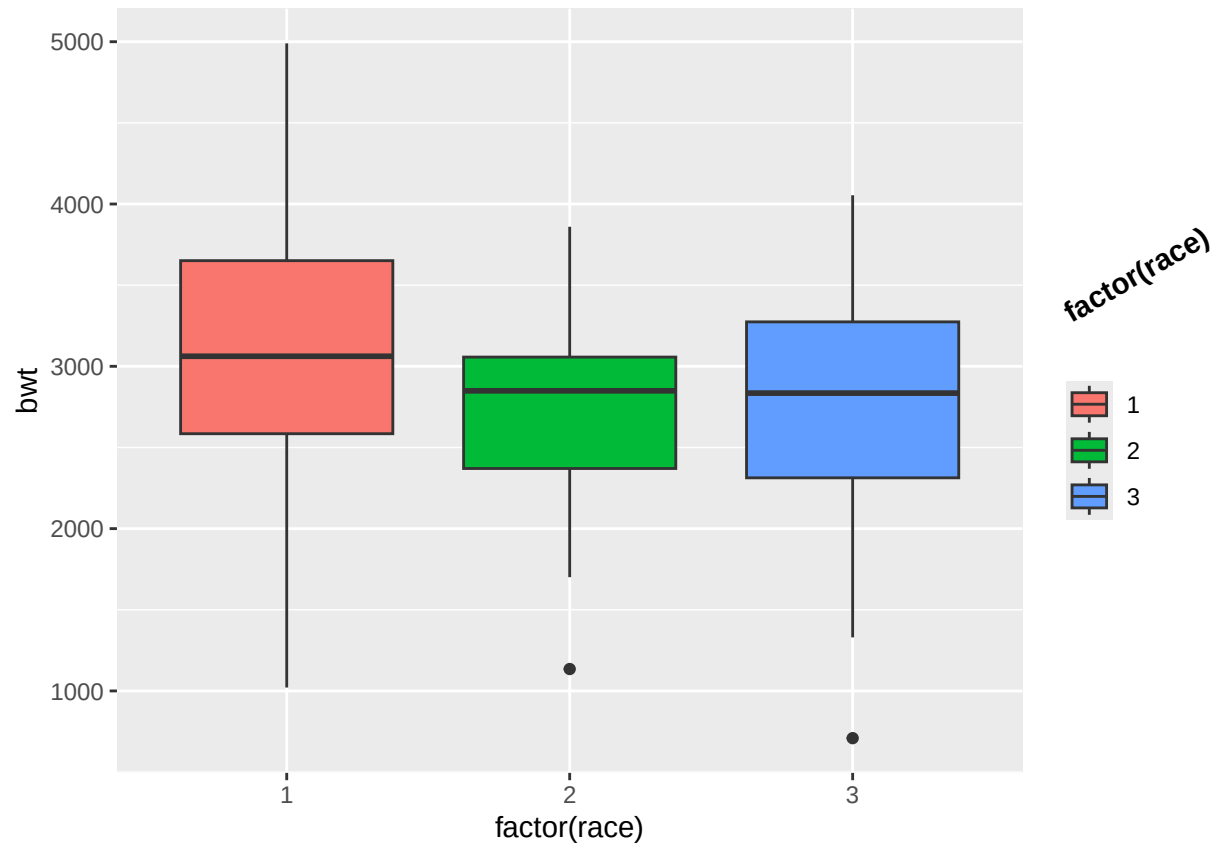
Legend

Appearance

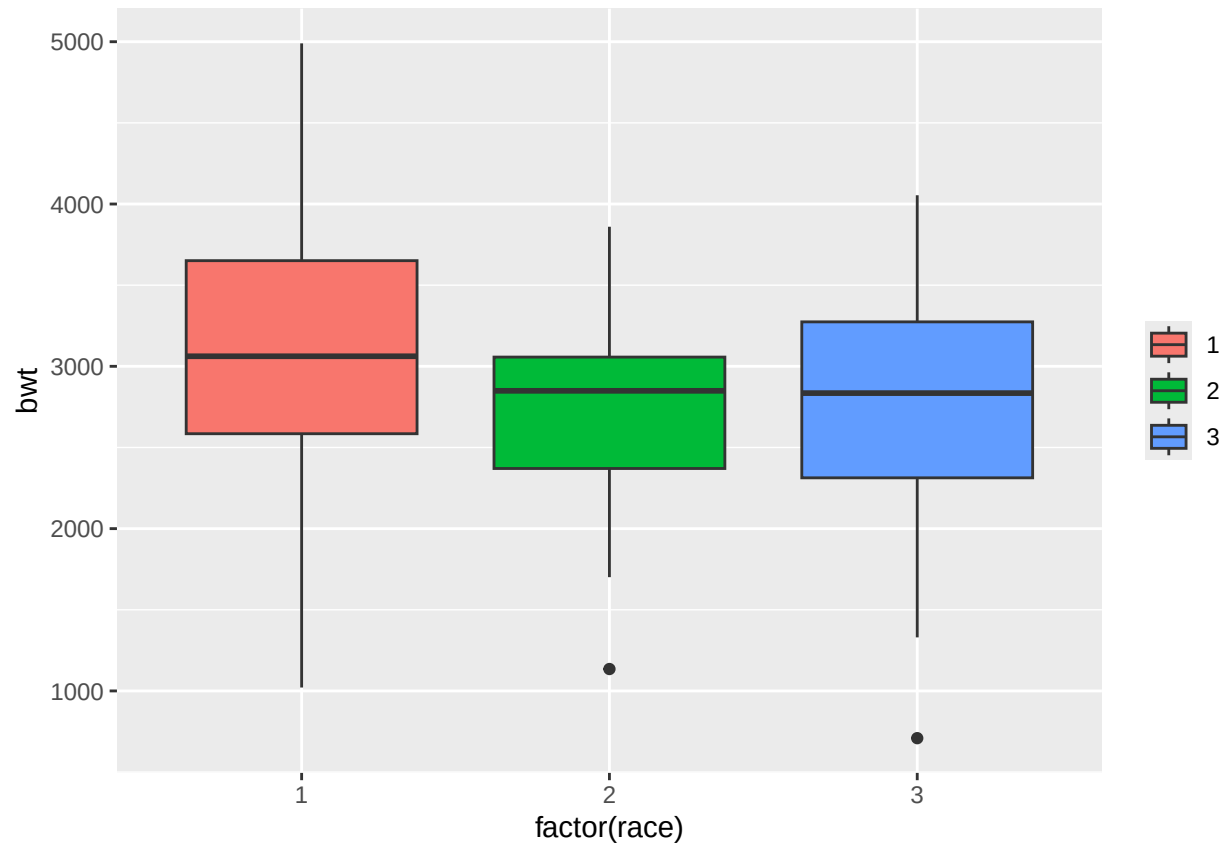
```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt,fill=factor(race)))
gg+geom_boxplot()+theme(legend.title=element_text
(face="bold",colour="seagreen",size=10))
```



```
gg+geom_boxplot()+guides(fill=guide_legend(title.theme=
element_text(face="bold",angle=30)))
```

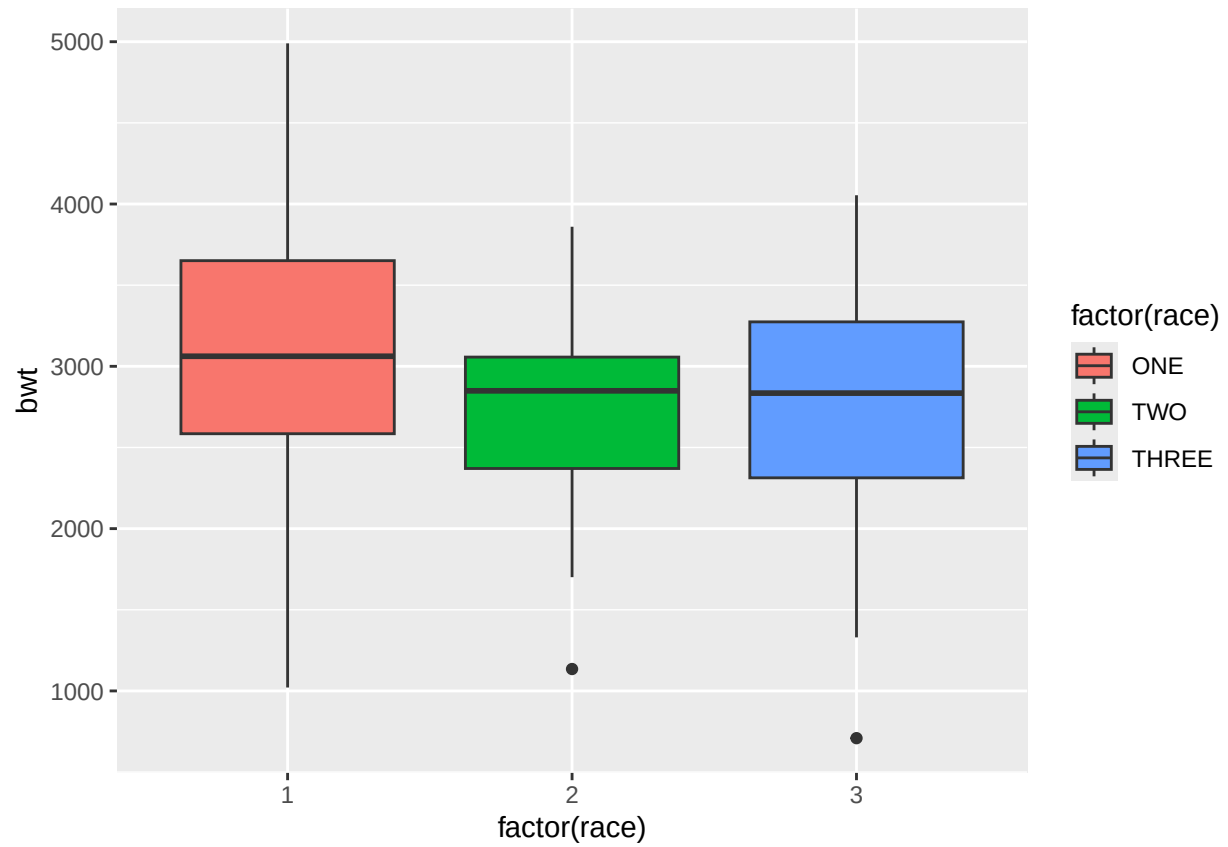


```
gg+geom_boxplot()+guides(fill=guide_legend(title=NULL))
```

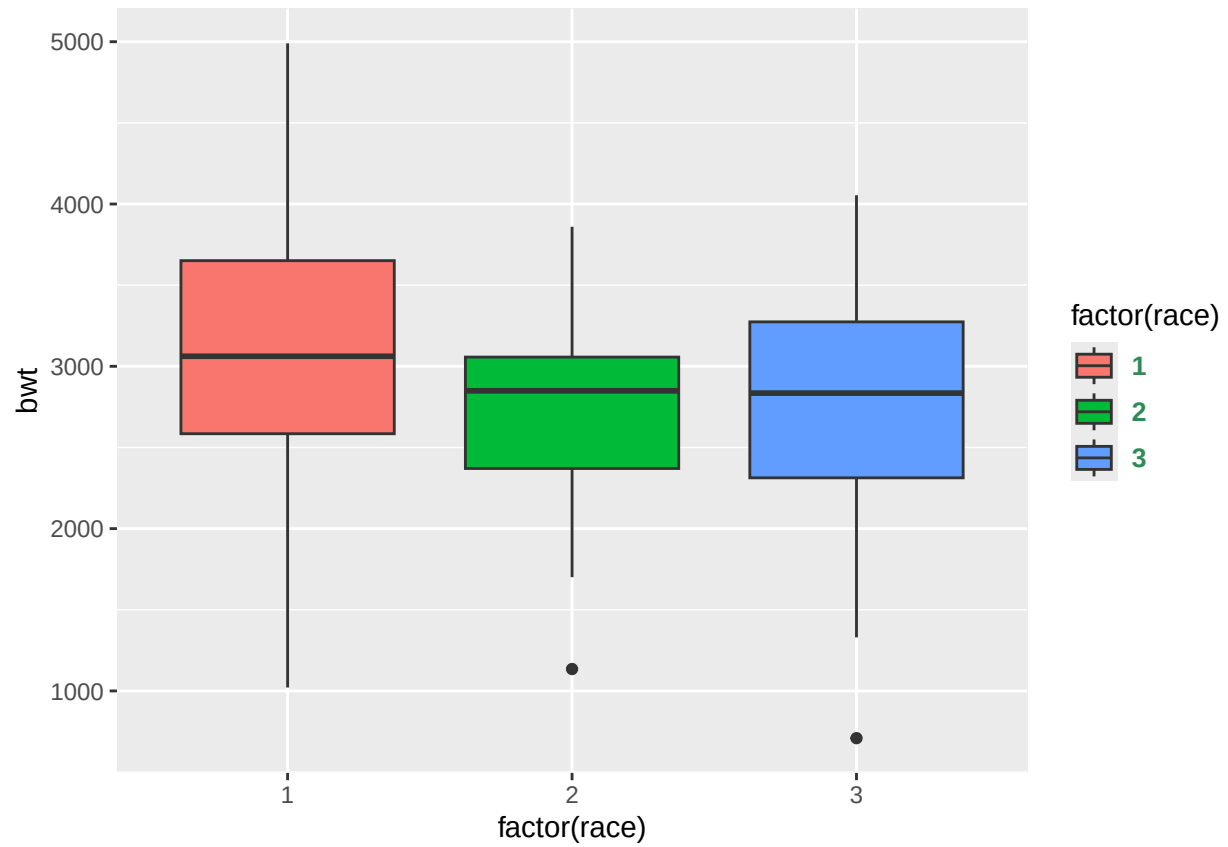


Labels

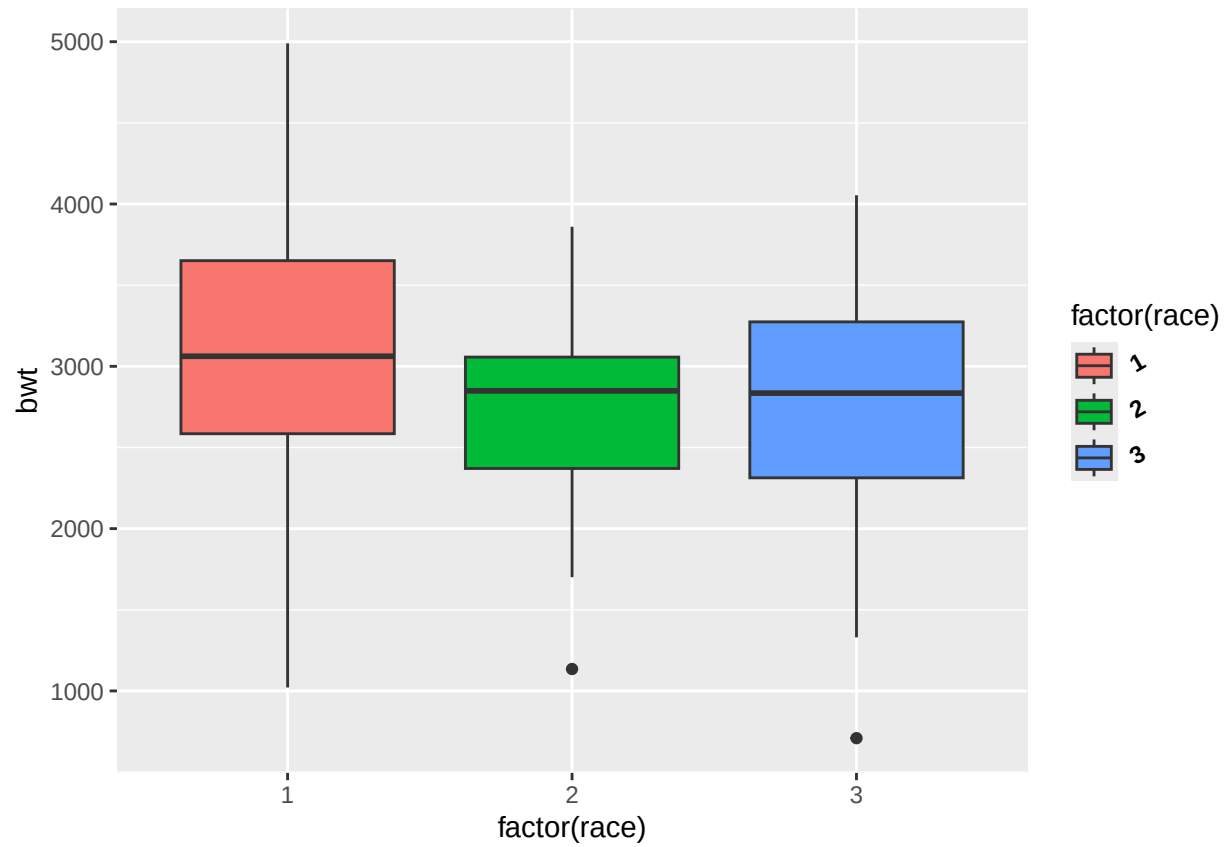
```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt,fill=factor(race)))
gg+geom_boxplot()+
scale_fill_discrete(labels=c("ONE","TWO","THREE"))
```



```
gg+geom_boxplot()+  
theme(legend.text=element_text(face="bold",colour="seagreen",size=10))
```

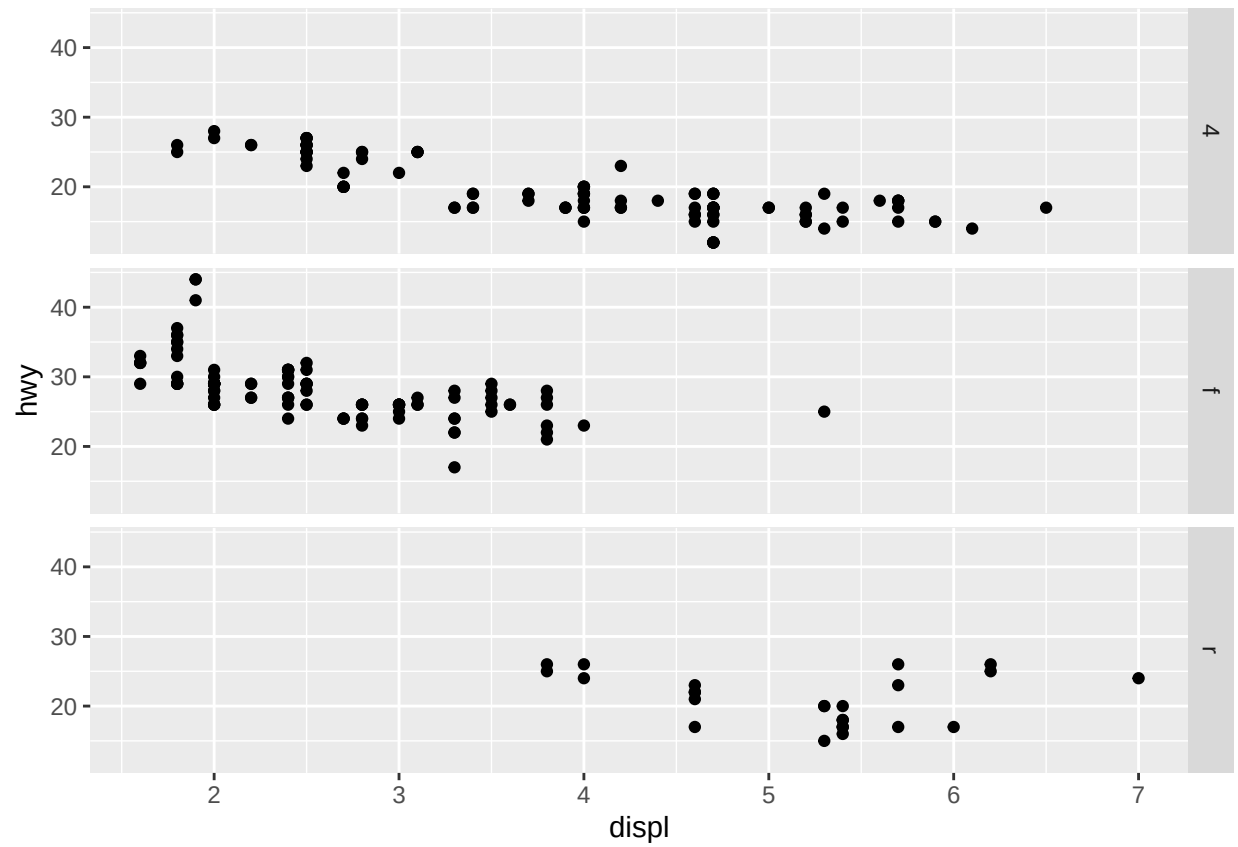



```
gg+geom_boxplot()+guides(fill=guide_legend(label.theme=
element_text(face="bold",angle=30)))
```

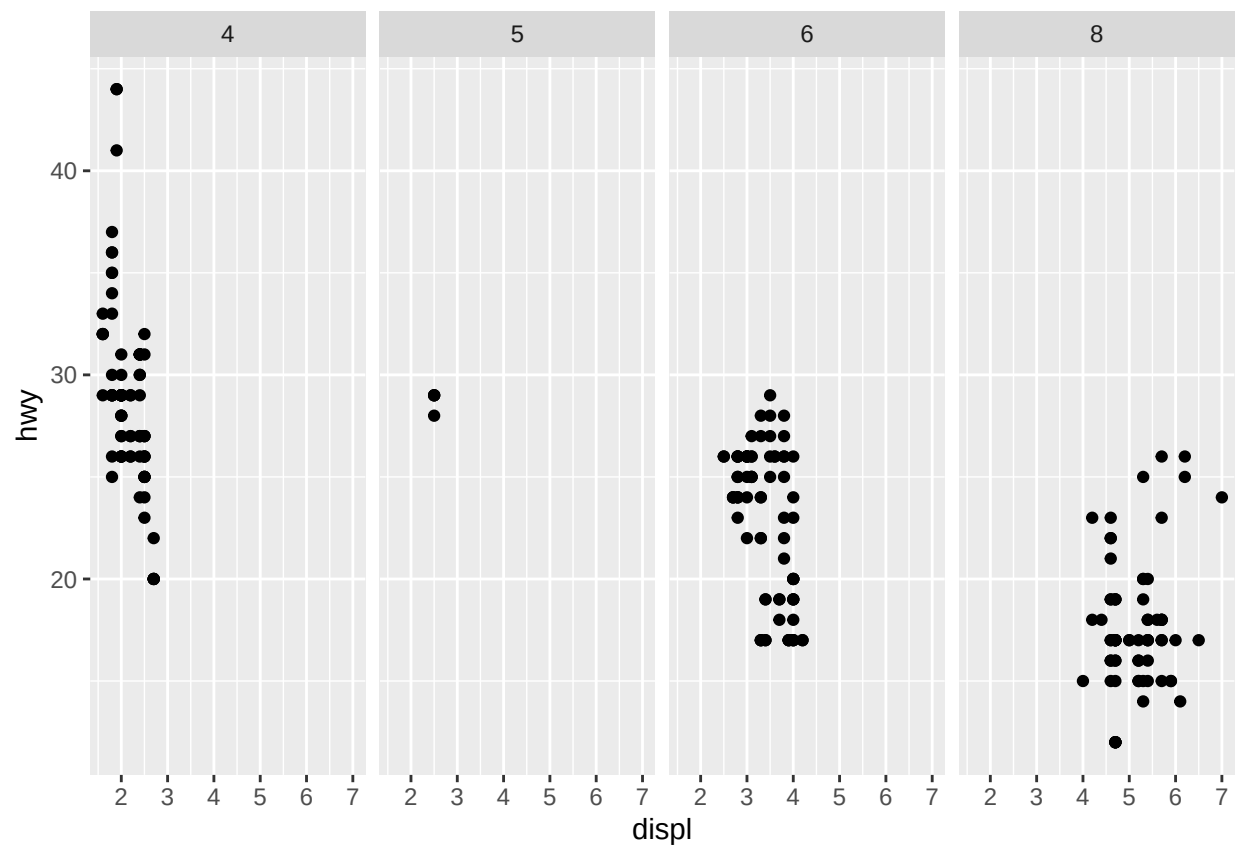


Facets

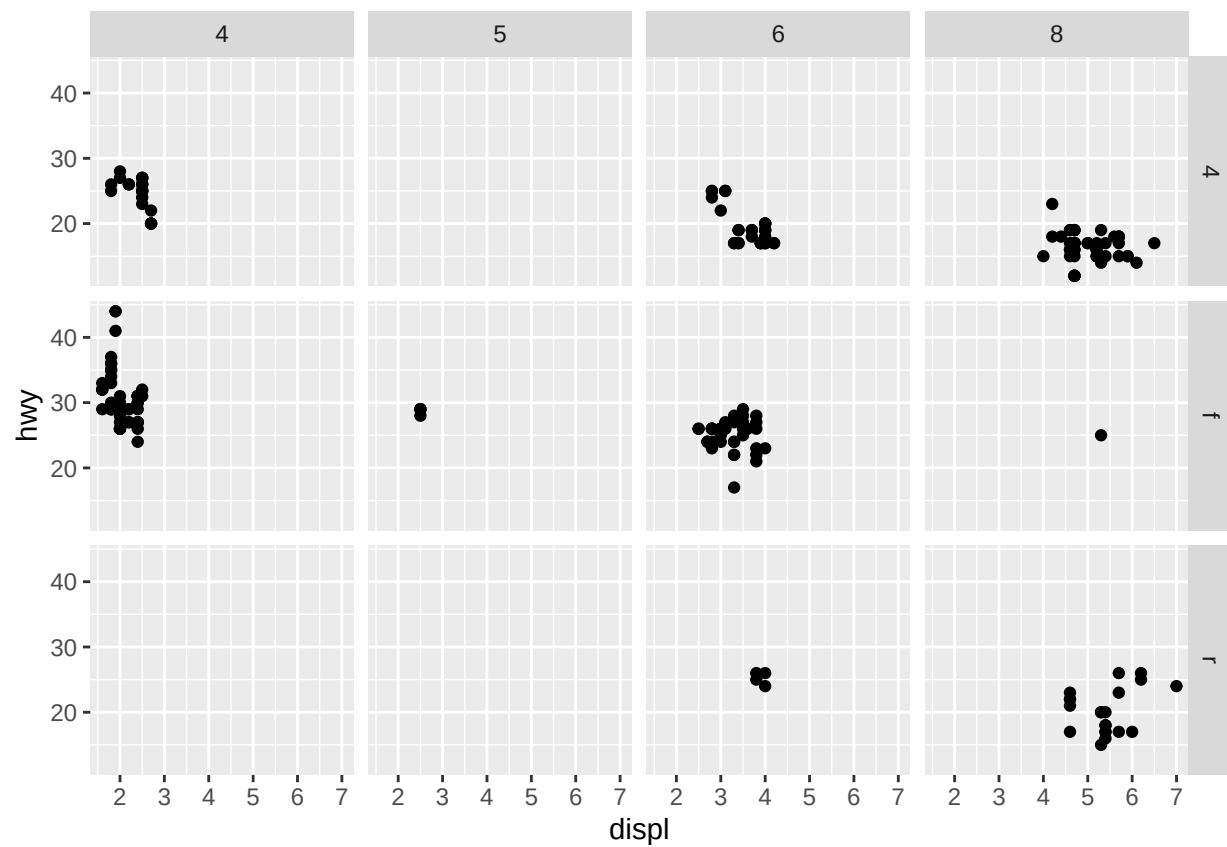
```
gf<-ggplot(mpg,aes(x=displ,y=hwy))+geom_point()  
gf+facet_grid(drv~.) #vertically
```



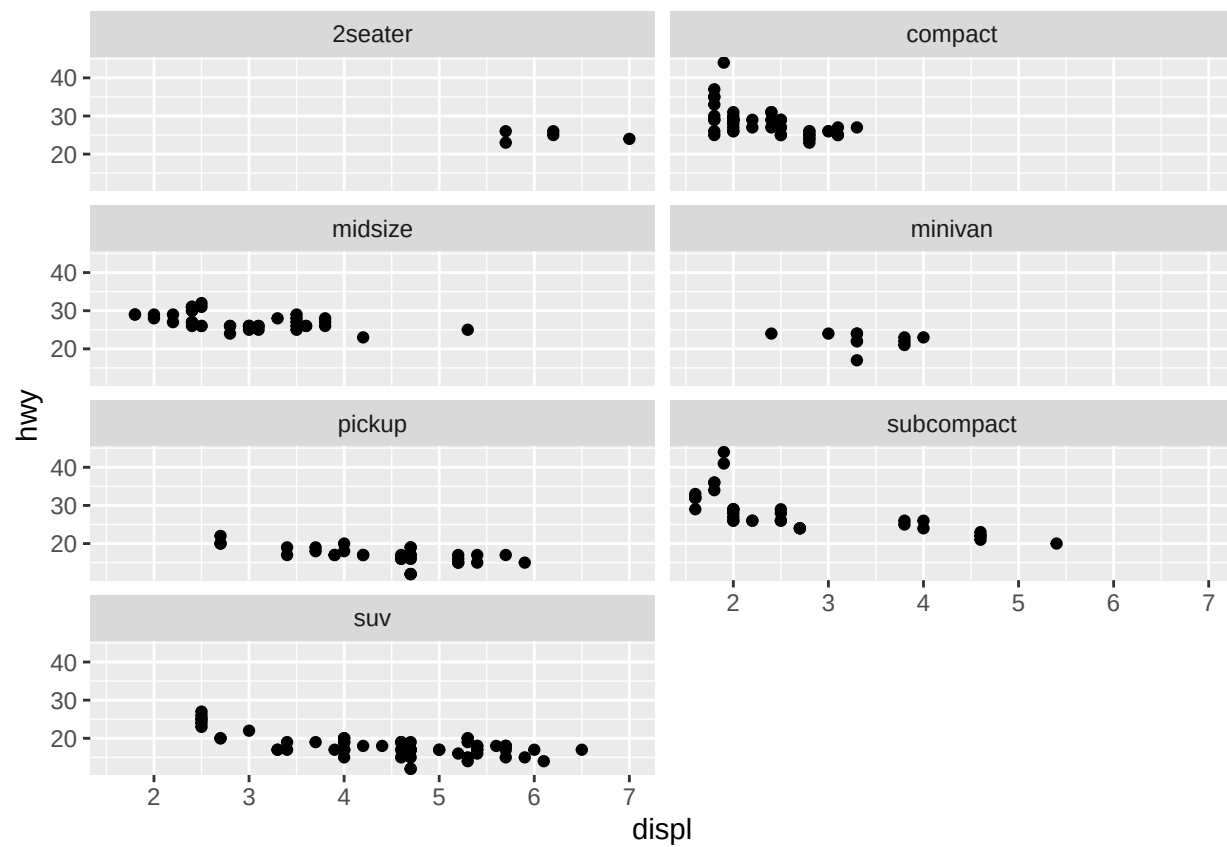
```
gf+facet_grid(.~cyl) #horizontally
```



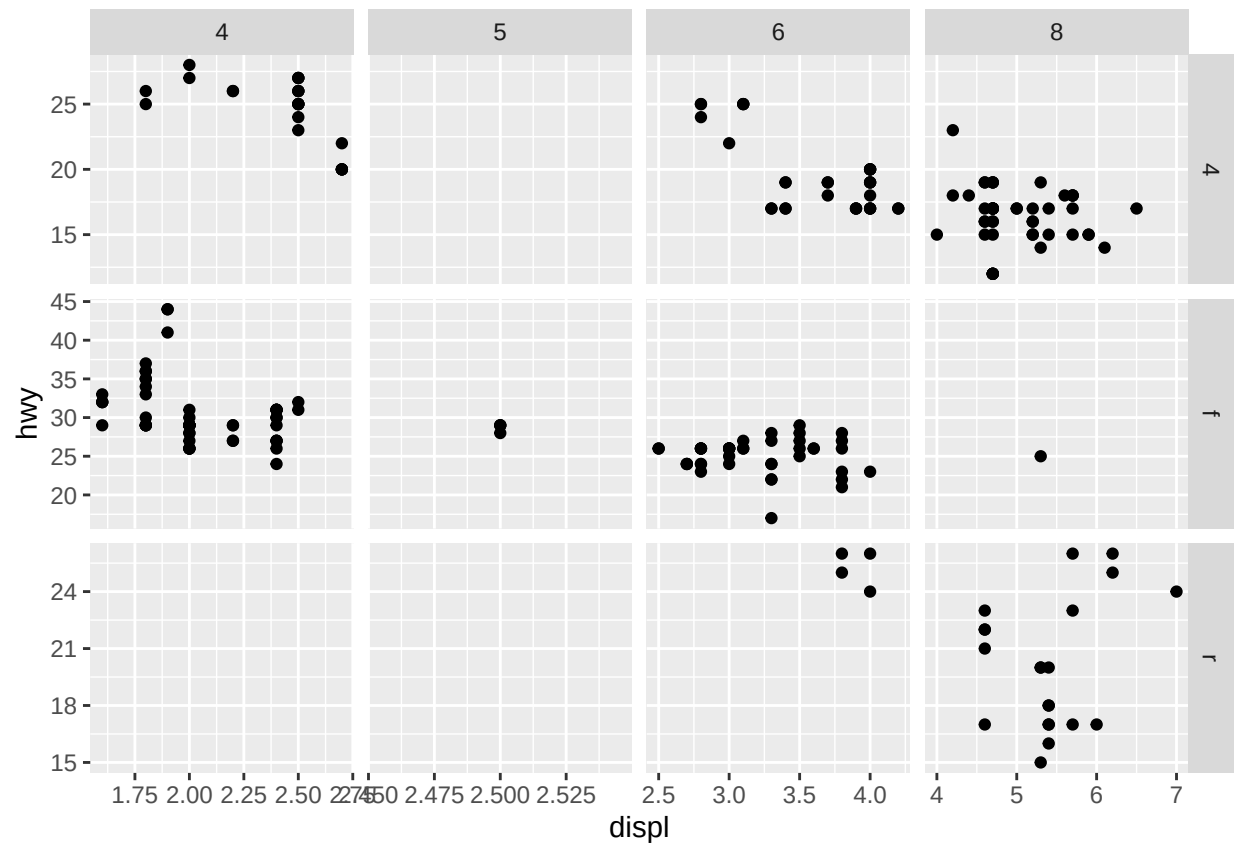
```
gf+facet_grid(drv~cyl)
```



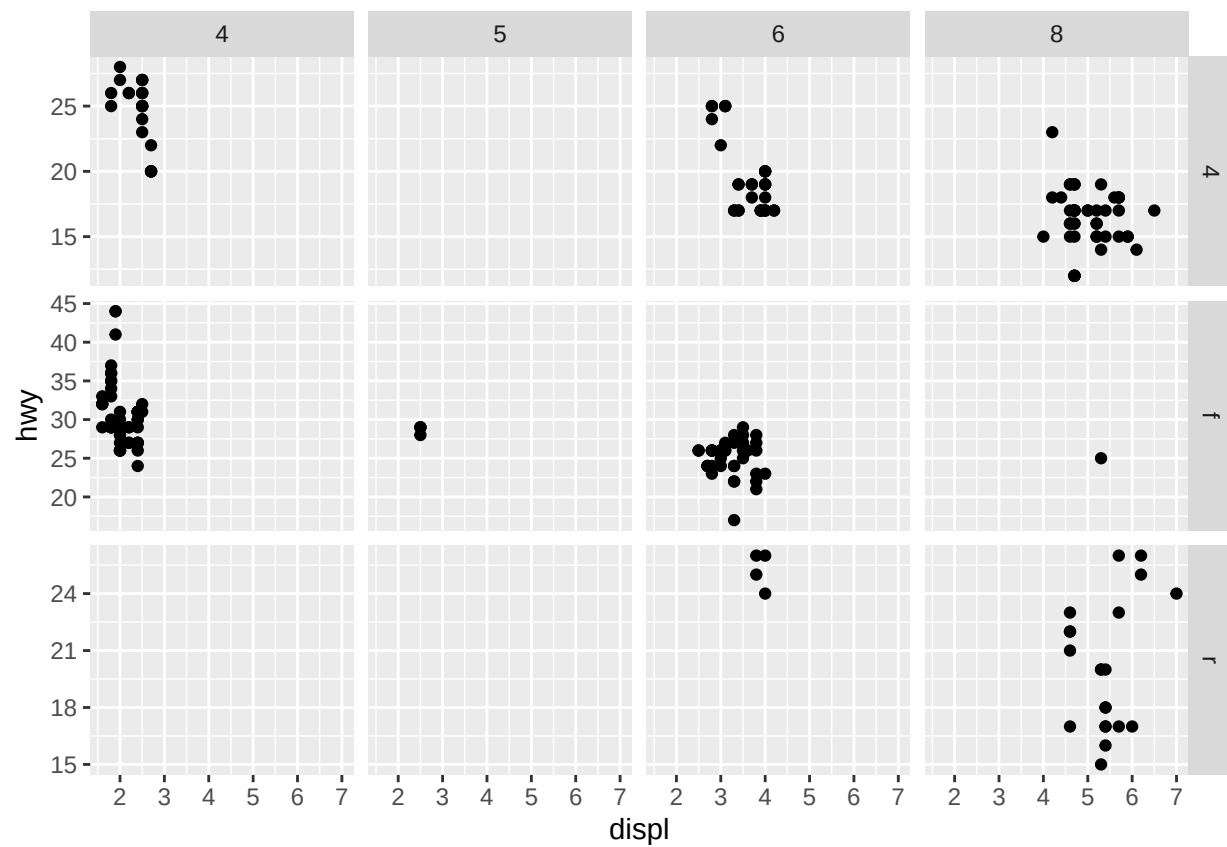
```
gf+facet_wrap(~class,ncol=2)
```



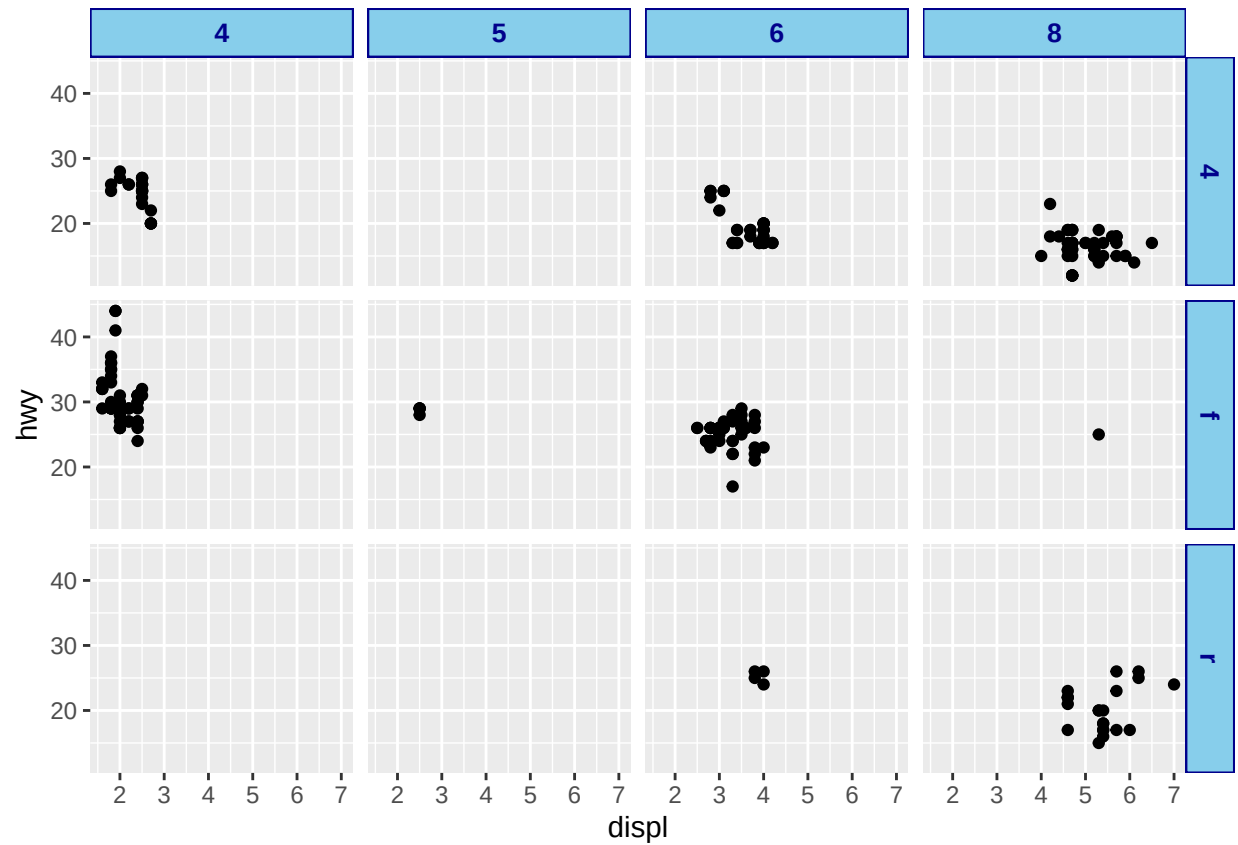
```
gf+facet_grid(drv~cyl,scales="free")
```



```
gf+facet_grid(drv~cyl,scales="free_y")
```

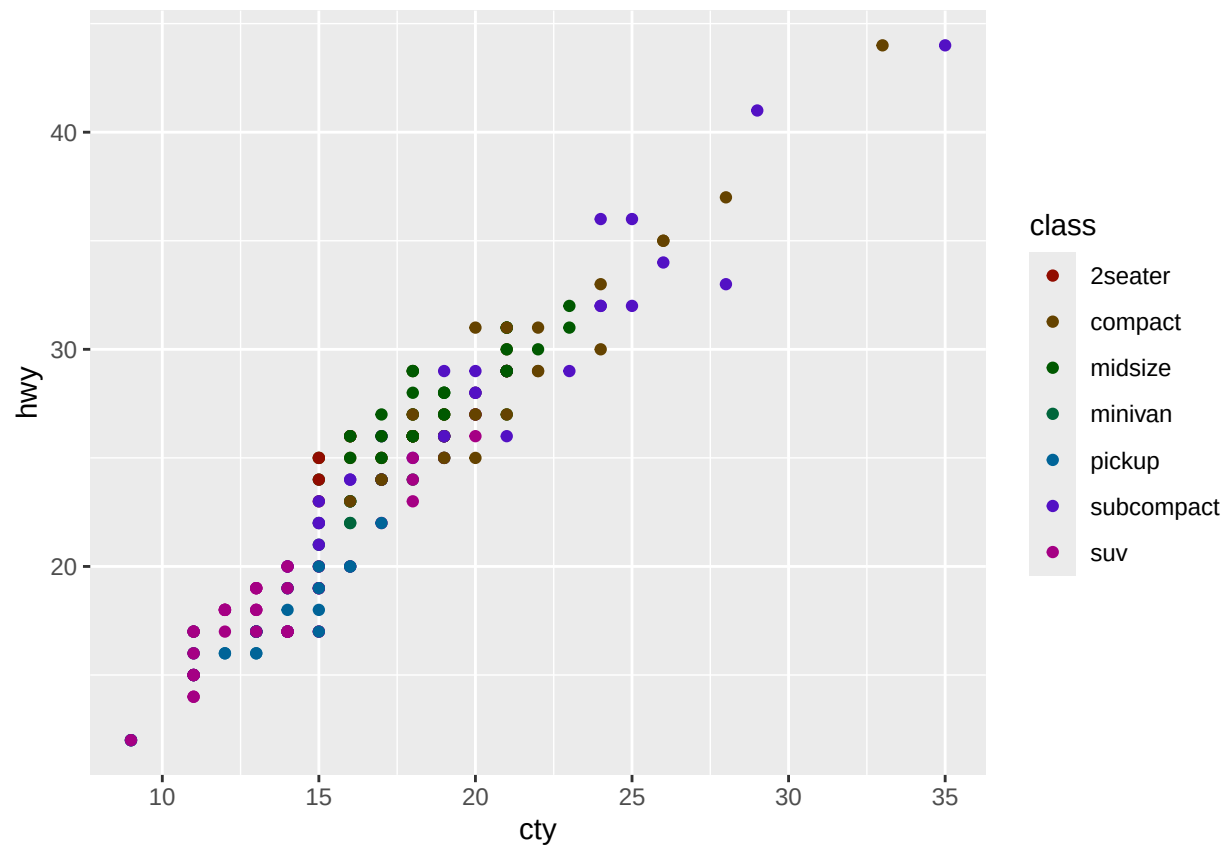


```
gf+facet_grid(drv~cyl)+
  theme(strip.text=element_text(face="bold",colour="darkblue",size=10),
        strip.background=element_rect(fill="skyblue",colour="darkblue"))
```

Colour Palettes

```
ggplot(mpg,aes(x=cty,y=hwy,colour=class))+geom_point()+
scale_colour_hue(l=30)
```



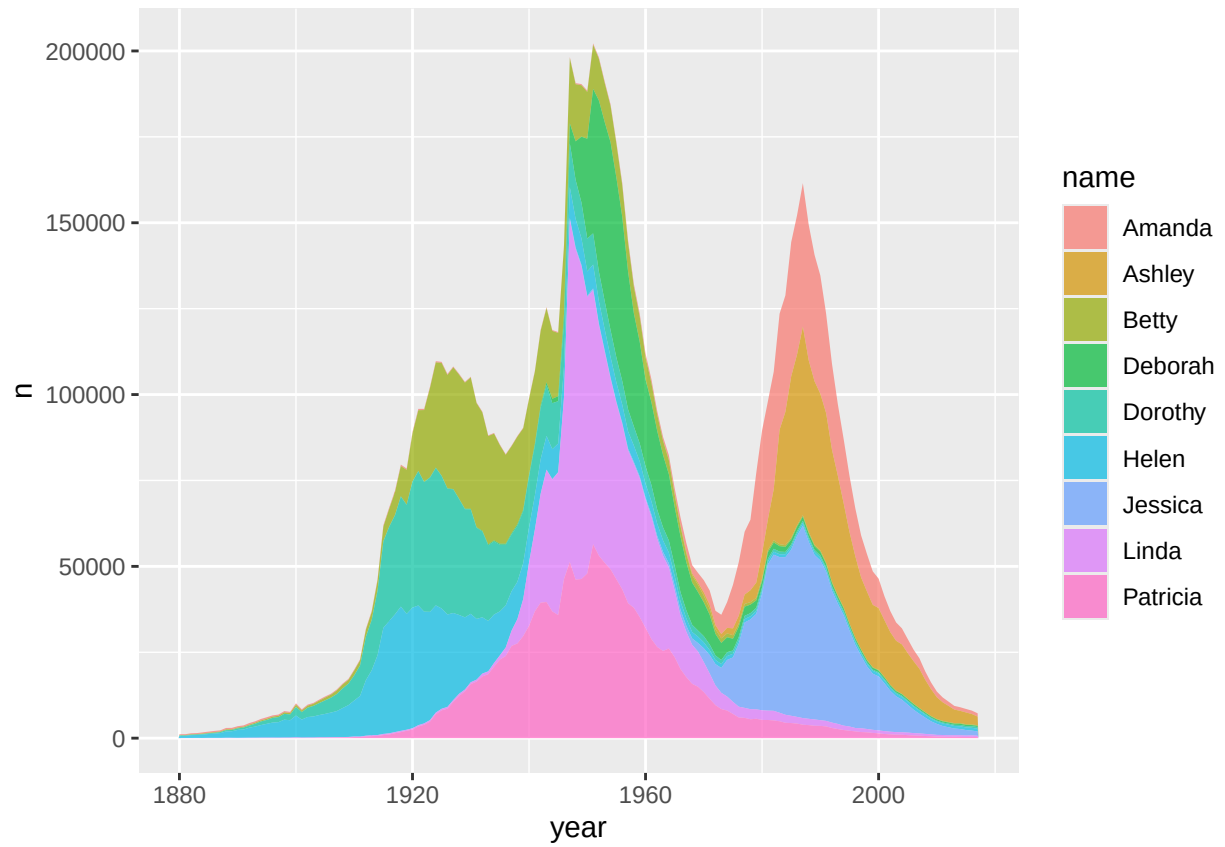
```
library(babynames)
```

```
## Warning: package 'babynames' was built under R version 4.4.3
```

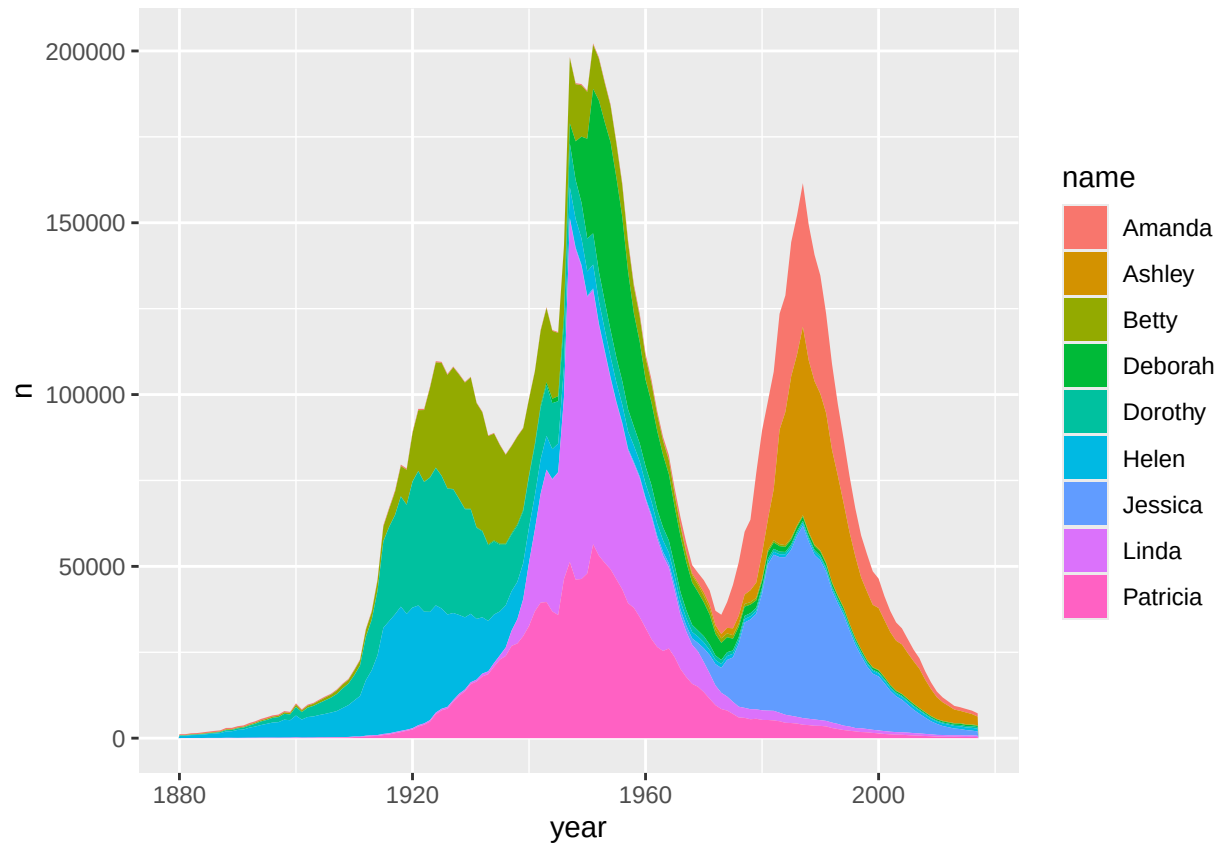
```

baby<-subset(babynames,subset=(sex=="F"&
name %in% c("Ashley","Amanda", "Jessica","Patricia","Linda","Deborah","Dorothy","Betty","Helen")))
gs<-ggplot(baby,aes(x=year,y=n,fill=name))
gs+geom_area(alpha=0.7)

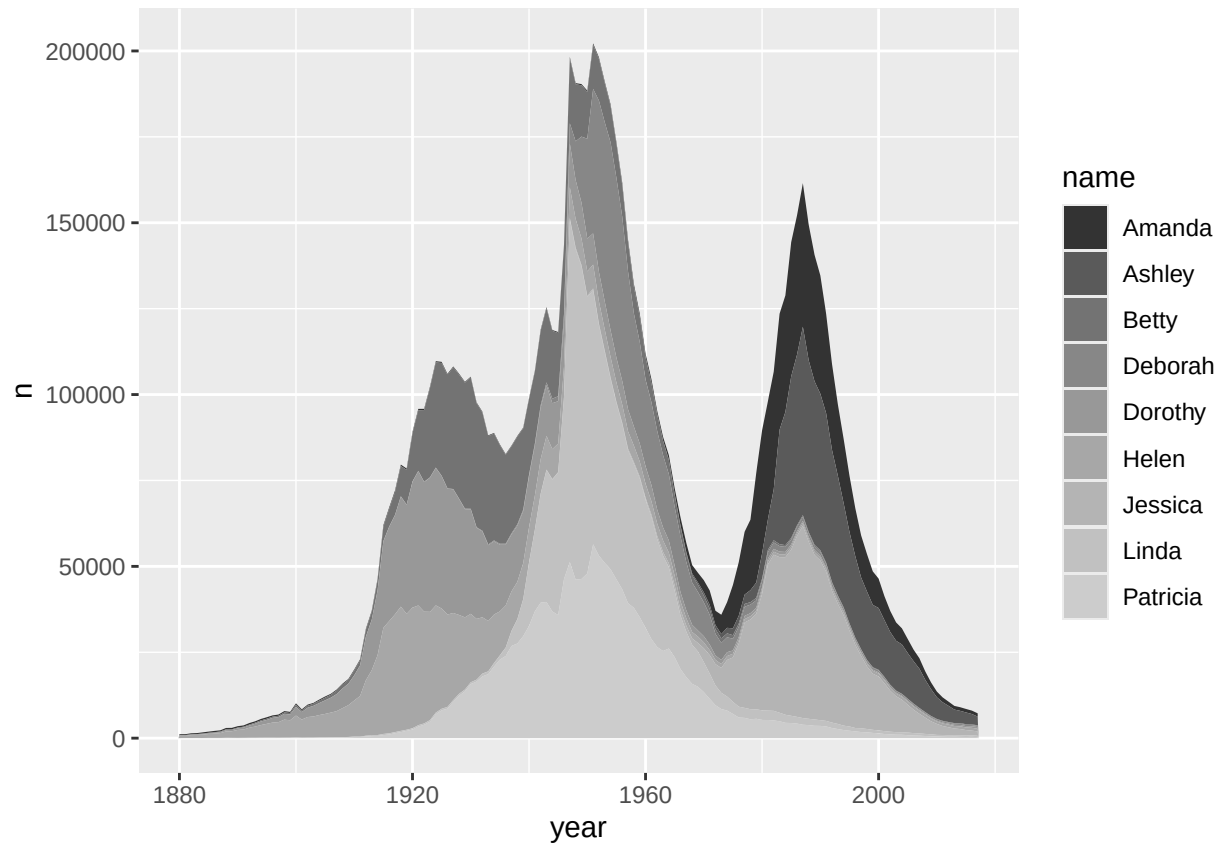
```



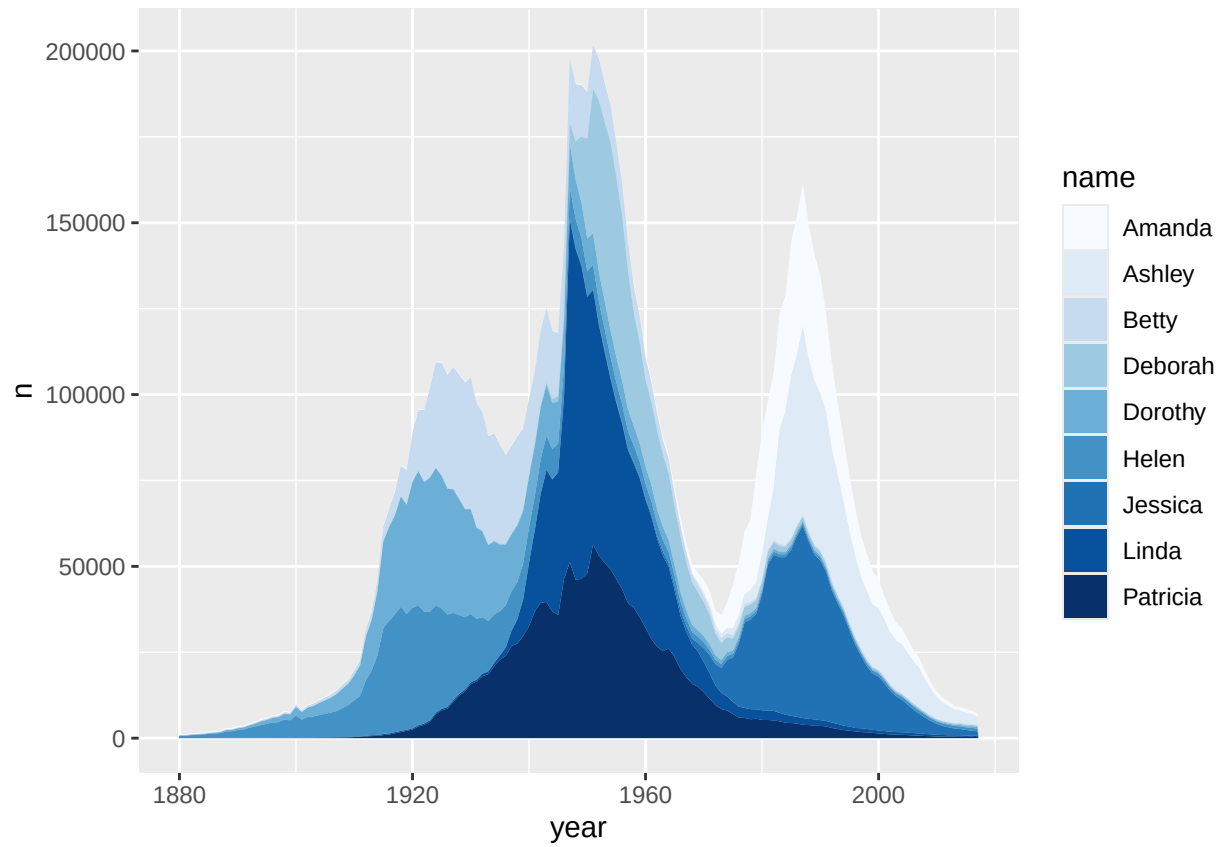
```
gs+geom_area()+scale_fill_discrete()
```



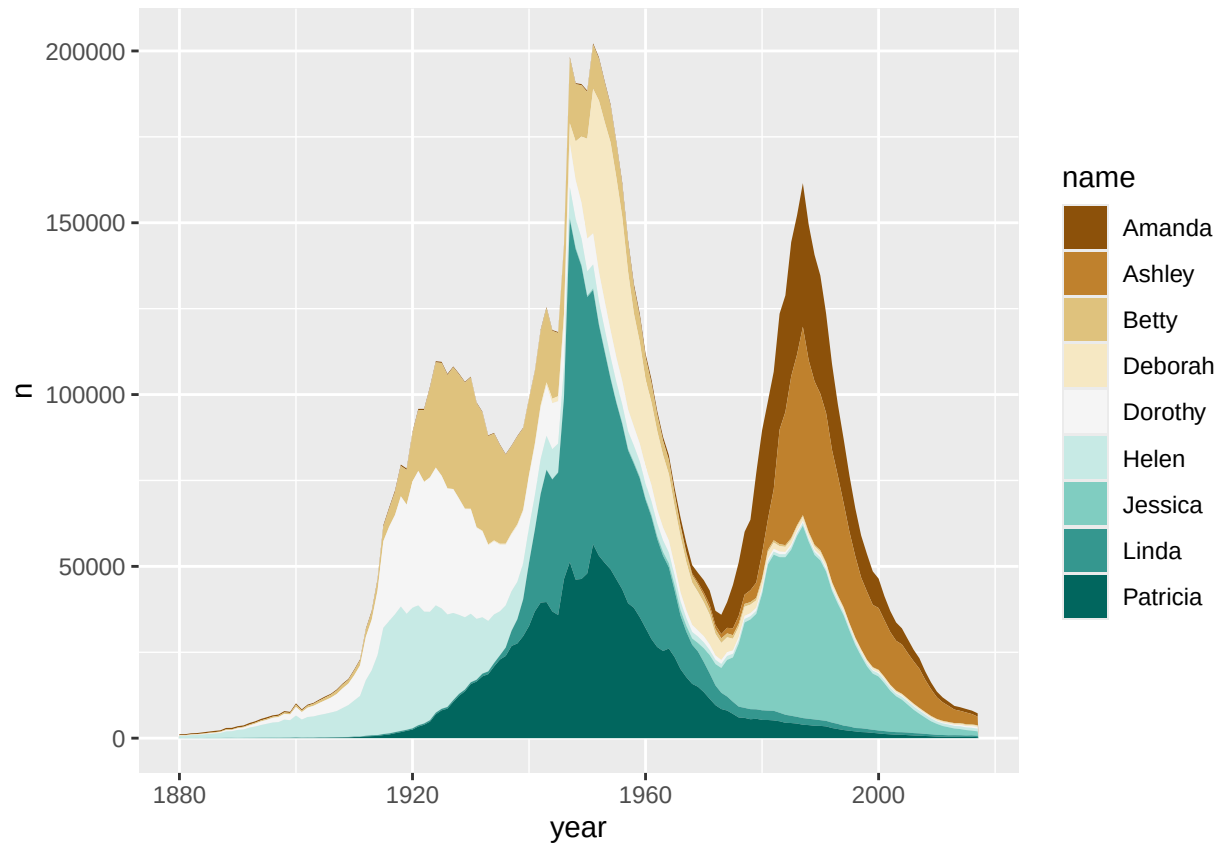
```
gs+geom_area()+scale_fill_grey()
```



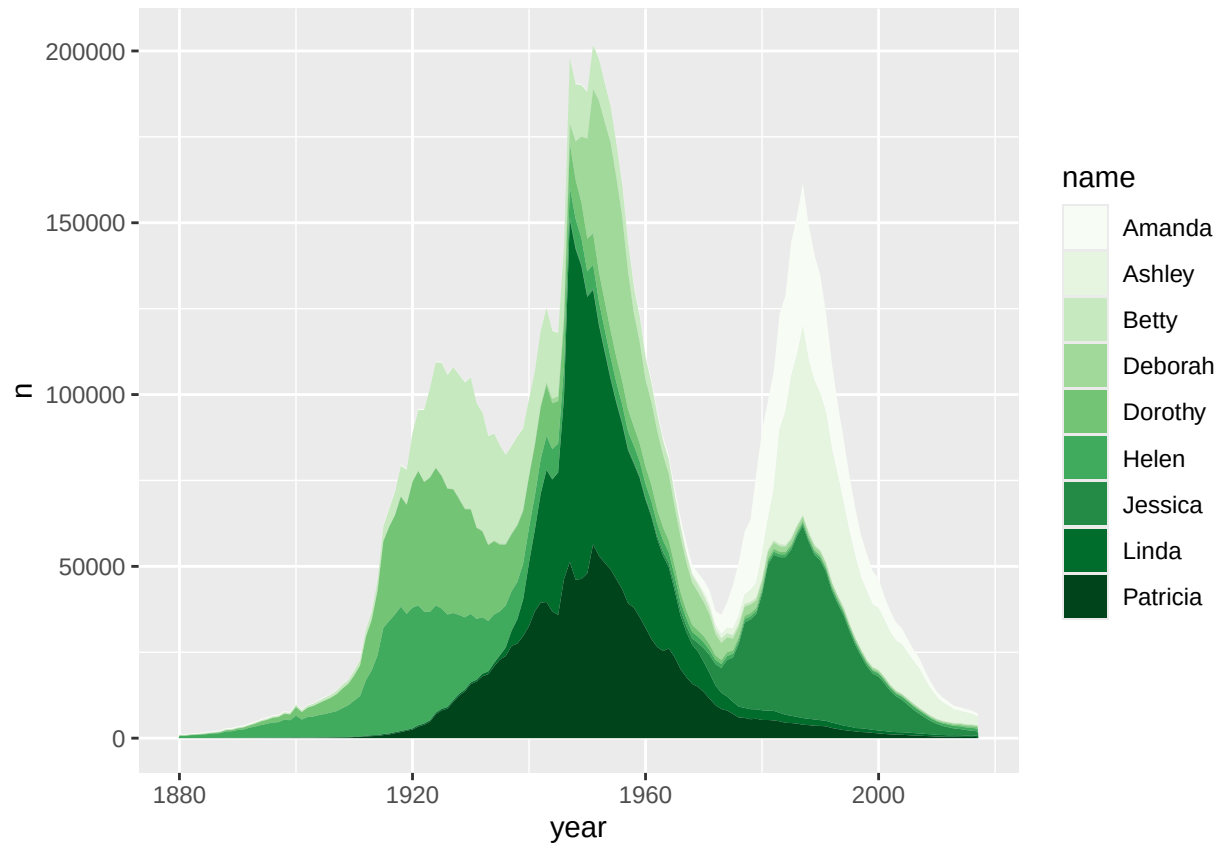
```
gs+geom_area()+scale_fill_brewer()
```



```
gs+geom_area()+scale_fill_brewer(palette="BrBG")
```

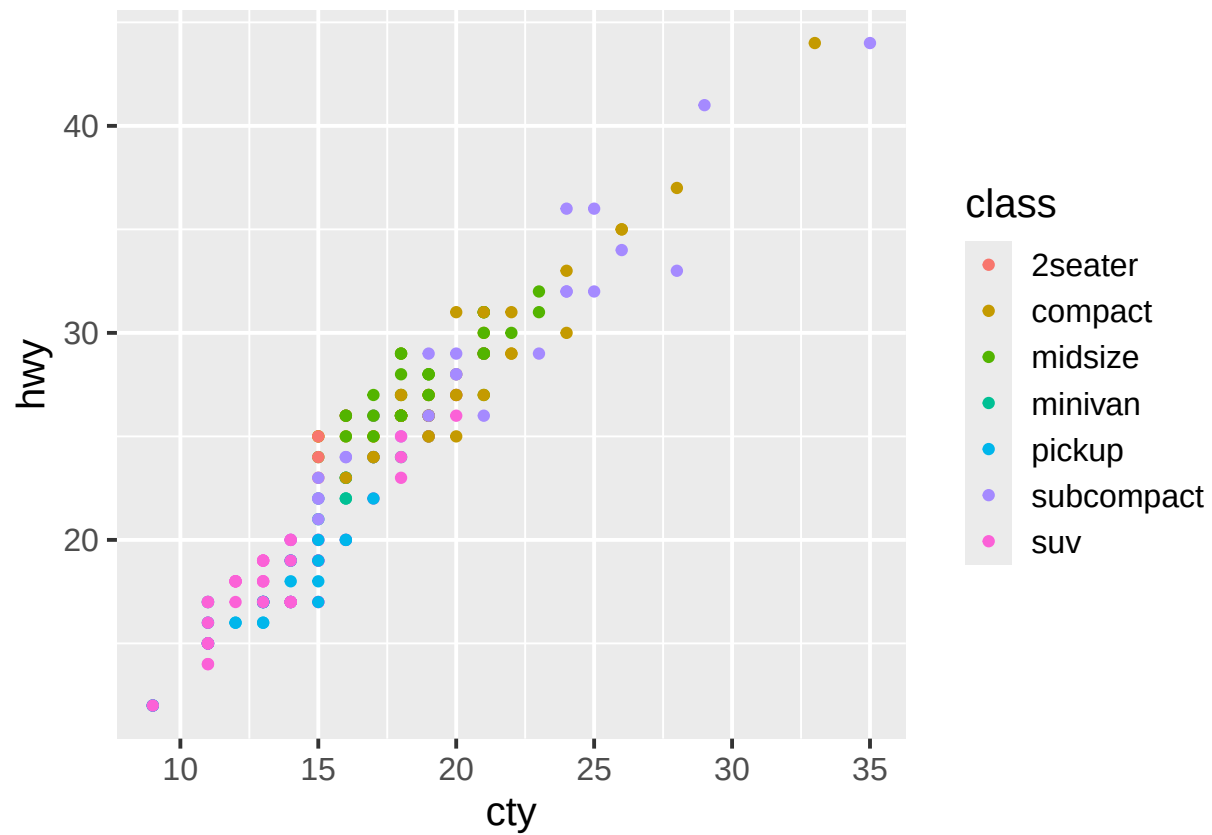


```
mypal<-brewer.pal(9,"Greens")
gs+geom_area()+scale_fill_manual(values=mypal)
```

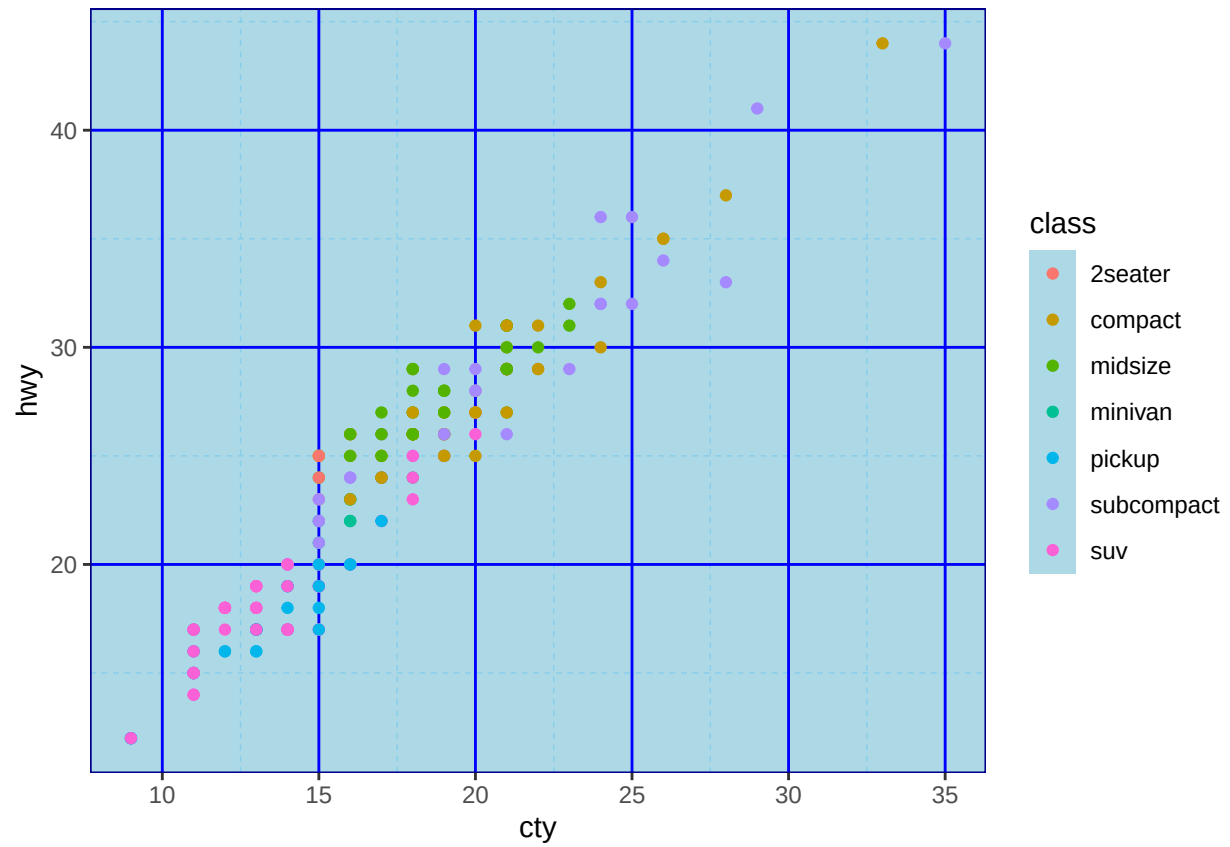


Themes

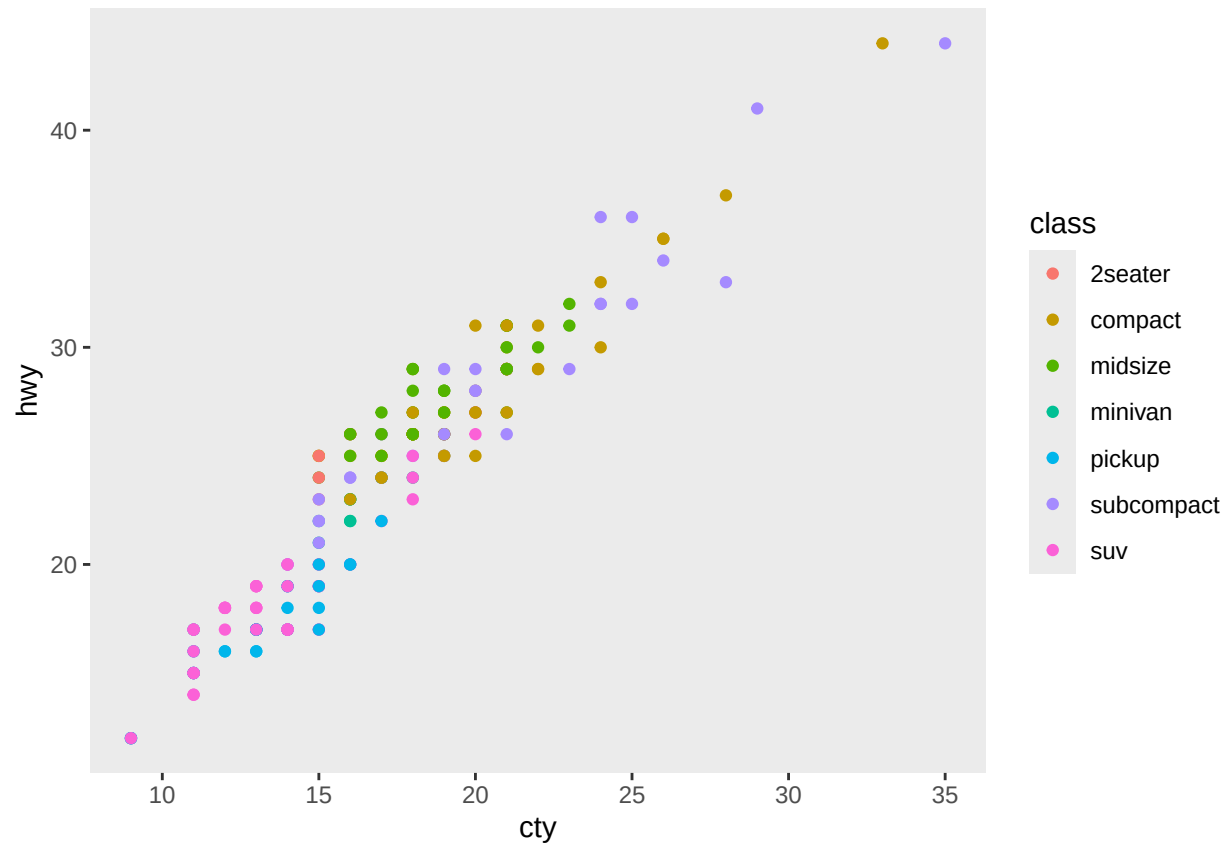
```
ggplot(mpg, aes(x=cty, y=hwy, colour=class)) + geom_point() + theme_bw()
```

```
ggplot(mpg,aes(x=cty,y=hwy,colour=class))+geom_point()+
theme(panel.grid.major=element_line(colour="blue"),
panel.grid.minor=element_line(colour="skyblue",linetype="dashed"),
panel.background=element_rect(fill="lightblue"),
panel.border=element_rect(colour="darkblue",fill=NA))
```



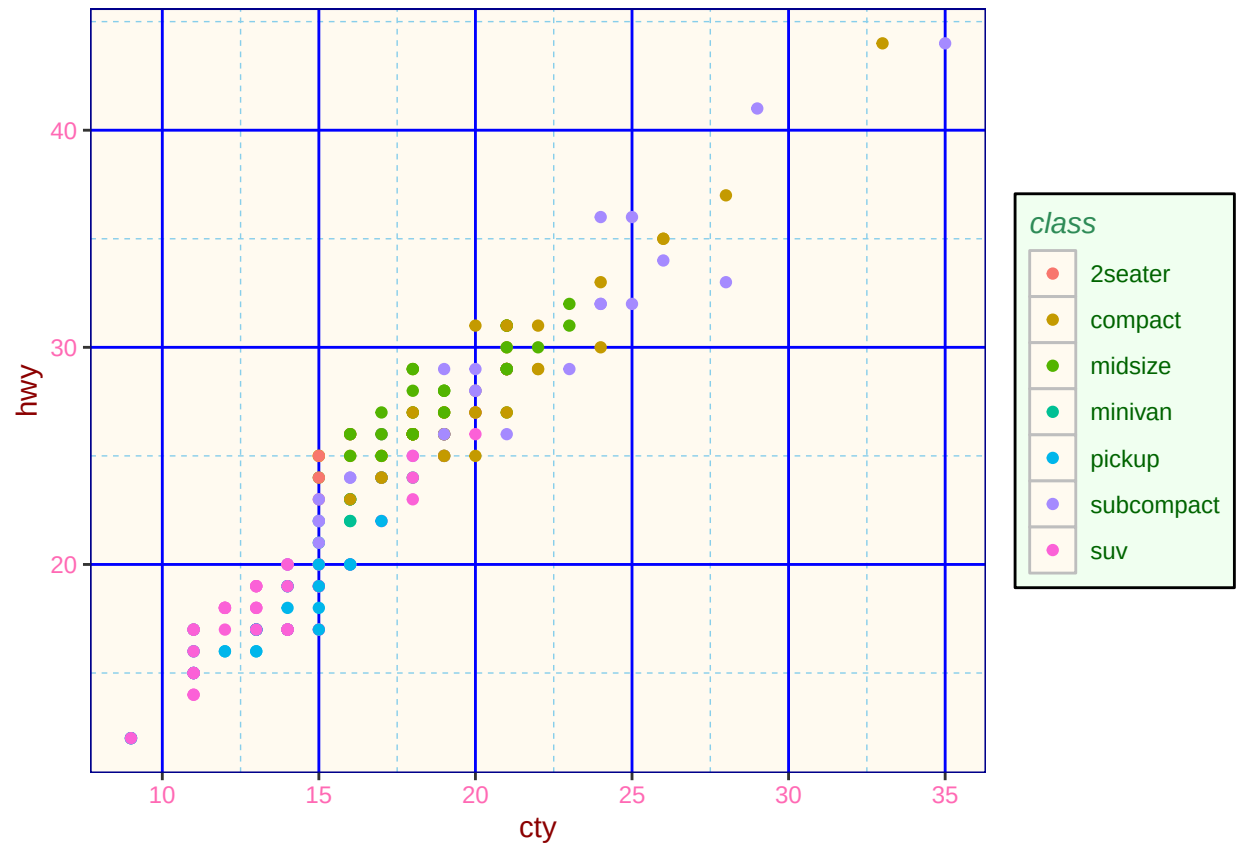
```
ggplot(mpg,aes(x=cty,y=hwy,colour=class))+geom_point()+  
theme(panel.grid.major=element_blank(),  
panel.grid.minor=element_blank())
```



Themes

```
mytheme<-theme_bw()+
theme(panel.grid.major=element_line(colour="blue"),
panel.grid.minor=element_line(colour="skyblue",linetype="dashed"),
panel.background=element_rect(fill="floralwhite"),
panel.border=element_rect(colour="darkblue",fill=NA))+
theme(axis.title.x=element_text(colour="darkred"),
axis.text.x=element_text(colour="hotpink"),
axis.title.y=element_text(colour="darkred"),
axis.text.y=element_text(colour="hotpink"),
plot.title=element_text(colour="lightcoral",face="bold"))+
theme(legend.background=element_rect(fill="honeydew",colour="black"),
legend.title=element_text(colour="seagreen",face="italic"),
legend.text=element_text(colour="darkgreen"),
legend.key=element_rect(colour="grey"))

ggplot(mpg,aes(x=cty,y=hwy,colour=class))+geom_point()+mytheme
```



```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt,fill=factor(race)))
gg+geom_boxplot()+mytheme
```

